

Research Proposal for a One-Medium Brane–Bulk Toy Physics Model

A Far-Field-First Test of Light, Forces, and Particle-Like Throats

Status: Classical toy-model research proposal.

The proposal organizes a falsifiable research program. It does not claim that the real universe is literally constituted this way or that any open target has already been derived.

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0.2 – Lean Skeleton

Document Control

Document role	Plain-language ontology explanation and top-down research proposal. It defines the questions, sequence, cross-sector handoffs, and decision points; it is not the canonical derivation or status ledger.
Canonical definitions	Imported from the current ontology and closure ledger. The proposal does not silently redefine symbols, ontological commitments, epistemic status, or failure conditions.
Detailed companion plans	The guided-light/photon plan, opposite-orientation throat plan, and cross-sector compatibility review retain their detailed derivation programs.
Current calculation status	Controlled by the derivation ledger and an independent Codex repository audit, not by a dated “Current State of the Program” chapter.
Old proposal material	The earlier 40-chapter skeleton and Chapters 0–10 are reference-only source material. They are not assembly inputs unless a passage is deliberately rewritten into the lean structure.
Structure control	Eight numbered chapters. No additional chapter or large appendix is added without an explicit structural revision.
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Scope control

Core drafting rule. Every substantial paragraph in the proposal should help answer at least one of these questions: What is the model? What must a sector reproduce? What calculation will test it? What does it require from the common medium? How will those requirements be combined? What work follows after the far-field program? Detailed governance, dated status, exhaustive provenance, and reusable templates belong in separate project artifacts.

Provisional Abstract

This proposal defines a falsifiable research program for a classical one-medium brane–bulk toy analog. The model assumes preferred time and four Euclidean spatial dimensions. Observed three-dimensional space is identified with an ordered, finite-thickness, shear-supporting brane within a larger compressible medium; the brane and bulk are ordered and de-structured states of the same globally conserved substance rather than separate materials. Light is sought as a guided transverse brane excitation with two polarizations, while particles are hypothesized to be finite oriented throats supported in part by a trapped transverse standing mode. Throat orientation supplies the proposed electric sign. Localized one-way throat drainage, throat-region pressure, stress, and order conversion, the separate distributed reordering of bulk material into ordered brane material represented by the positive bulk-to-brane contribution to S_{leak} , and the resulting reservoir response are proposed as the material basis for a calibrated gravity observable. Motion and acceleration of the same complete dressed configuration are intended to produce magnetism, radiation, passive response, and inertia.

The central question is whether one mathematically coherent medium can support all required sectors simultaneously without independent retuning. The research therefore begins with a far-field-first capability analysis rather than with a premature nonlinear particle solve. Guided light and the stronger localized photon-packet hypothesis are tested alongside gravity, static electricity, magnetism, radiation from accelerating charge, inertia, drag, leakage, and unwanted modes. Each calculation must determine what behavior a brane observer would see, what mechanism and response operator could produce it, and what fields, coefficients, characteristic speeds, source conventions, conservation laws, and exclusions it imposes on the shared medium. The resulting requirements are then intersected rather than judged as separately tuned successes.

If that intersection is nonempty, the surviving field content, dynamical branch, constitutive terms, parameters, boundary data, constraints, and reservoir variables will be frozen as Candidate Parent System V1. The candidate must support a stable finite brane and complete background spectrum, after which all far-field sectors are rederived from that same system and parameter region. Only a candidate that survives this common-medium test advances to the nonlinear program: self-consistent supported throats, orientation partners and species structure, moving-throat response, two-body forces, finite-thickness contact and possible annihilation or neutral binding, and the fixed point between localized one-way throat drainage, the global bulk reservoir, and the separate distributed S_{leak} return across the porous brane.

Success would establish a controlled classical analog architecture within its stated scope, not a derivation of the real universe, exact general relativity, exact electromagnetism, or quantum theory. An empty compatibility region, an unavoidable extra mode or loss channel, failure of the supported-

throat branch, or inability to close the global conservation and reservoir ledgers would be an equally legitimate result because it would identify where the proposed architecture fails.

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Executive Summary

1.1 The model and its central test

This proposal defines a falsifiable research program for a classical toy analog. It does not claim that the real universe is literally constituted in this way or assume that its open targets have already been derived. The postulated arena has preferred time and four Euclidean spatial dimensions. One globally conserved compressible medium occupies two material states: an ordered, finite-thickness, shear-supporting brane-like state and a de-structured bulk-like state. The brane and bulk are therefore not separate substances; all transport, conversion, return, radiation, and reservoir response must ultimately close within the same medium.

Light is sought as the lowest gapless transverse branch guided by the ordered brane, with two freely selectable polarizations. This conditional linear-light sector is distinct from the stronger nonlinear hypothesis of a finite-energy, three-dimensionally localized classical photon packet. Particles are hypothesized to be finite nonlinear throats through the brane. A trapped transverse standing mode may help hold a throat open, but its support energy is structural and is not identified automatically with observed particle mass. The side of the brane toward which a throat is oriented supplies the proposed electric sign, while particle species and a complete antiparticle map require additional internal, support-mode, topological, or composite structure.

The other observable sectors are intended to arise from the response of the same medium. Localized one-way throat drainage, throat-region pressure, stress, and order conversion, the separate distributed porous-brane reordering encoded by the positive bulk-to-brane contribution to S_{leak} , and reservoir response are proposed as the source-side material basis for a calibrated gravity observable. Motion of an oriented throat is expected to produce a sign-dependent magnetic response, while acceleration must be tested for emission into both transverse light polarizations. Inertia is attributed to reconfiguration of the complete dressed object: its throat geometry, trapped mode, near fields, flow, conversion region, and surrounding medium. Active gravitational strength, passive gravitational response, and inertial resistance are kept distinct until their signs and low-energy universality are derived.

The central question is whether one mathematically coherent parent medium can support all mandatory sectors in compatible regimes with one field content, one dynamical branch, one coefficient set, and one closed treatment of boundary and reservoir data. If Λ_S is the admissible region left by sector S , the decisive requirement is

$$\Lambda_{\text{common}} = \bigcap_{S \in \mathcal{S}_{\text{required}}} \Lambda_S \neq \emptyset.$$

A successful light point, electric point, and gravity point selected from incompatible branches or parameter regions would not satisfy this test.

1.2 The far-field-first research program

Because the final parent equations and nonlinear throats are not yet closed, the program begins with far-field capability requirements rather than allowing an unresolved particle core to supply whatever source a later calculation needs. For each sector, the proposal first states what a brane observer must see, identifies the proposed material mechanism, specifies the static and dynamic calculations needed to test it, and extracts what the result requires or forbids in the common medium. Requirement discovery works backward from observable behavior; mathematical closure later works forward from one frozen parent system to its background, spectrum, sources, and nonlinear solutions.

Light leads because it directly tests the host medium. The first track asks whether a stable finite slab supports a guided, gapless transverse branch with two polarizations, acceptable confinement across the normal direction, and controlled dispersion, leakage, and mixing. The second asks whether finite-thickness dispersion and a slaved reflection-even material dressing can maintain a localized photon-packet candidate without collapse, strong generic interaction, or unacceptable radiation. Failure of the packet track would not erase a valid conditional classical light-wave sector, but it would reject the intended self-bound free-photon interpretation.

Gravity asks whether localized one-way throat flow, stress, and order conversion, together with the separate distributed porous-brane S_{leak} return and the resulting reservoir response, can produce a probe-independent field with an attractive inverse-square local regime, orientation-even source behavior, one frozen calibration, acceptable dynamics, and calculable return-induced screening or overshoot. Static electricity asks whether the complete finite-slab response contains a healthy reflection-odd carrier with leading k^2 static stiffness, the required interaction sign, sufficiently universal throat coupling, and a conserved oriented count or current. Parity or gaplessness alone is not enough, and every additional long-range even or odd mode must be included.

Magnetism and radiation then test whether electricity belongs to one dynamical source hierarchy rather than remaining an isolated static analogy. The same oriented source and coupling must govern the leading static electric response, the velocity-order magnetic response, and acceleration-order emission into both transverse light polarizations. Radiation into odd, even, longitudinal, bulk, and order-conversion or reservoir branches must be calculated rather than omitted. The far-field phase also establishes requirements for positive passive and inertial response, retarded passivity, low-energy species universality, compatible characteristic speeds, and acceptable leakage, emission, friction, and drag.

Each sector hands its required fields, operators, speeds, source conventions, normalizations, conservation laws, permitted departures, and failure conditions to a common Medium Capability Matrix. This exposes shared quantities and conflicts before sector-specific repair can hide them.

1.3 From sector requirements to one common medium

The sector outputs are normalized and intersected. A mandatory failure cannot be compensated by strong performance in another sector, and the union of independently successful parameter points is not a common solution. The surviving region must be reported honestly as empty, isolated, fragile, disconnected, or robust. The compatibility tests are potential falsification gates, not established contradictions.

If a nonempty region remains, the project will freeze Candidate Parent System V1. The freeze must specify the arena, field inventory, conservative, relaxational, or mixed dynamical branch, equations and constraints, operator basis and derivative order, symmetries, independent parameters, boundary and reservoir data, source and observable conventions, and terms forbidden from later sector-specific insertion. A coefficient selected to localize a photon must also propagate through the slab spectrum, electric response, throat support, radiation, and drag. A coupling fixed by static electricity must retain its role in magnetism and acceleration radiation. Any substantive revision creates a new candidate version and reopens its actual downstream dependencies.

Candidate V1 must then produce a finite, selected, stable or sufficiently metastable brane and its complete constrained spectrum. All far-field sectors must be rederived from that same background and common parameter region. Only this bottom-up rederivation can show that the capability contracts are realized rather than merely compatible on paper. The proposal supplies the top-down contract; the derivation ledger supplies the executable evidence, assumptions, frozen inputs, and dependency record used to decide it.

1.4 Beyond the far field and meaning of success or failure

Only a candidate that survives the common-background and far-field rederivation advances to the nonlinear particle program. The first task is to continue a localized transverse bound-mode seed or acceptably long-lived resonance on a provisional throat geometry into a self-consistent supported throat whose shape, support stress, transport, conversion, and reservoir loading agree. Its stability, source profiles, and couplings to the complete background spectrum must then be established.

Both orientations of each species must be solved from the same equations. Electric orientation, species identity, and antiparticle identity remain separate: a complete antiparticle candidate may require transformation of internal phase, circulation, chirality, or topology in addition to geometric reflection. Moving and accelerating throat families must determine inertia, passive response, magnetism, radiation, drag, and any low-energy universality among active, passive, and inertial roles. Two-throat calculations must continue from the far field through the finite-thickness crossover and into full nonlinear core geometry. Projected coincidence, cancellation of the leading odd electric disturbance, four-dimensional core contact, and annihilation are distinct events. Same-species throat-antithroat annihilation must therefore be distinguished from possible neutral binding of oppositely charged objects with different internal or composite structure.

Finally, the local throat and global medium must be solved as one feedback problem. Localized one-way throat drainage and throat-region order conversion alter the bulk reservoir; material does not return through the throat. Instead, because the finite-thickness brane is porous, bulk material

can reorganize into the ordered brane state across it. For a projected brane observer, that distributed bulk-to-brane return is the positive contribution to the signed term S_{leak} . It changes the brane and the asymptotic data seen by the throat. Closure requires a fixed point with consistent material, momentum, energy, and, where relevant, entropy accounting. A local throat solution is not complete if its own sustained outflow would invalidate the boundary data used to obtain it.

The possible outcomes are deliberately layered. A valid linear-light sector is a result even if the localized photon hypothesis fails. Individually viable far-field sectors are useful but do not establish one common medium. Failure of Candidate V1 rejects that candidate, not automatically every allowed realization of the broader architecture. The architecture fails only if its mandatory capabilities prove mutually incompatible throughout the admitted frozen theory class or require prohibited retuning or an unsupported mechanism.

Full success requires one controlled parent system to survive the common far-field intersection, the stable-brane and complete-spectrum tests, the supported-throat program, moving and interacting solutions, and global reservoir closure. Even that result would establish a classical analog architecture only within its stated scope, not exact general relativity, exact electromagnetism, quantum theory, or a derivation of the real universe. An empty common region, unavoidable wrong-sign or extra long-range mode, excessive leakage or drag, failed photon-packet or throat branch, unacceptable species dependence, or unclosed reservoir cycle is an equally legitimate result because it identifies where the proposed architecture fails.

Plain-English Ontology

Ontology describes what the candidate model says exists and how its familiar phenomena are to be interpreted. It does not establish that the required solutions exist. Some elements below are defining postulates, some are physical interpretations of quantities that a brane observer would measure, and some are open hypotheses that later calculations may confirm or reject. The canonical ontology and closure ledger remains the authority for exact definitions, branch conditions, symbols, and failure criteria; this chapter supplies the readable mental picture needed to understand the research program.

The proposal is deliberately a one-medium construction. Light, particles, charge, gravity, magnetism, radiation, inertia, and return are not assigned to independent substances that can be tuned separately. They are intended to be different behaviors of one material system. The central scientific burden is therefore not merely to make each interpretation plausible in isolation, but to show that one frozen parent medium can realize all of them together.

2.1 Arena and one fundamental medium

The parent arena has one preferred time coordinate and four spatial coordinates. A brane observer labels an ordinary position by

$$\mathbf{x} = (x, y, z),$$

while a complete parent-space position is

$$\mathbf{X} = (\mathbf{x}, w).$$

The coordinate w is normal to the brane. The present candidate takes the four-dimensional spatial geometry to be flat and Euclidean, with preferred Newtonian time. This arena is a postulate of the toy model rather than a result to be derived. Changing it later would alter the meaning of distance, reflection, wave propagation, Green functions, and throat contact, so it would constitute a new architectural branch rather than a harmless change of notation.

A brane-bound observer normally records only $\mathbf{x}$. Two structures can therefore have the same

projected three-dimensional position while remaining separated in w . That distinction becomes physically important for finite throats: projected overlap of their apparent mouths does not by itself imply that their full four-dimensional cores touch.

The brane is presently assumed to be globally two-sided and co-orientable. In plain language, a continuous normal direction can be chosen throughout the brane, allowing the two sides to be compared consistently. Calling them $+w$ and $-w$ is conventional, but the availability of two distinguishable sides is part of the candidate geometry. The *sign* of the normal is used by the charge ontology, while the unsigned normal axis is enough to organize ordinary light guidance.

Everything in this arena is made from one compressible condensate. At the mean-field level it is represented schematically by

$$\psi = \sqrt{n} e^{i\theta},$$

where n is constituent density and the phase θ can generate material flow on a conservative branch. This field is a description of the substrate, not a claim that observed particles are collections of conventional massive constituents whose rest masses add up to the observed particle mass. The observed particle, if the model succeeds, is an emergent configuration of the medium.

Total constituent content obeys a parent conservation law,

$$\partial_t n + \nabla_4 \cdot \mathbf{J}_n = 0.$$

Material may move, compress, change its degree of order, and exchange energy among resolved and unresolved modes, but it may not be created from an empty bulk or disappear into an external supply. An apparent source or sink seen by a three-dimensional observer must ultimately be a projection of four-dimensional transport or a conversion between states of the same medium.

Because the arena includes a preferred material frame, Lorentz symmetry is not fundamental. It could emerge approximately only if the relevant long-wavelength branches acquire sufficiently aligned speeds and couplings. The speeds of transverse light, longitudinal brane motion, bulk sound, electric response, and gravity-related response are therefore distinct until the common-medium calculation proves an acceptable relation among them.

2.2 Ordered brane and de-structured bulk

An order variable χ_B distinguishes two material states. Where $\chi_B \simeq 1$, the medium is ordered, brane-like, and able to carry the resolved shear structure required by the model. Where $\chi_B \simeq 0$, the same material is de-structured and bulk-like. The brane and bulk are therefore not two substances. They are two states of one globally conserved substance, and their exchange routes are not symmetric. Ordered material may de-structure as a localized one-way throat drain carries it into the bulk, but it does not return through the throat. Because the finite-thickness brane is porous, bulk material can instead reorganize into the ordered brane state across the brane; for a projected observer, the positive bulk-to-brane part of that distributed exchange is encoded in S_{leak} .

Observed space is identified with a finite ordered slab centered on a local mid-surface. It is not

an infinitesimally thin mathematical sheet. Bulk-like material lies on both sides, and the slab has two interfaces separated by a characteristic thickness H_{br} . The local mid-surface is relational: translating the entire slab through a homogeneous parent medium changes the coordinate origin but not the physical state. What matters is displacement relative to the slab's own center and interfaces.

While the two interfaces remain single-valued surfaces over the brane, their displacements can be written as h_+ and h_- . Two combinations are useful:

$$h_m = \frac{h_+ + h_-}{2}, \quad h_t = \frac{h_+ - h_-}{2}.$$

The first is the leading mid-surface displacement. It reverses under reflection through the slab and is therefore reflection odd. The second changes the local half-thickness and is reflection even. These parity labels organize the far-field calculation, but they do not guarantee that either variable is a complete physical eigenmode. Density, order, phase, longitudinal motion, bulk response, and reservoir variables may mix with both.

The two-interface description is also limited to the graph regime. It is useful for the stable background, weak waves, far fields, and separated deformations. It cannot represent a throat after the ordered region develops a tube, overhang, neck, fold, reconnection, pinch-off, or other topology change. In that regime the full parent fields and localized core diagnostics must replace the interface coordinates.

The ordered slab is intended to support in-plane shear. That property is what makes transverse light possible. The de-structured bulk is not assumed to carry the same resolved shear branch, so the contrast between the two states may allow the slab to act as a waveguide. Finite thickness also supplies a material length scale that can affect dispersion, interface mixing, electric response, throat size, and the crossover from long-range behavior to nonlinear core physics. It is therefore a load-bearing part of the ontology rather than a decorative embedding picture.

The exact dynamics of order remain a major branch choice. In a reversible realization, changes in χ_B behave as inertial material dynamics and carry energy and momentum coherently. In a relaxational realization, loss and recovery of order may produce entropy and require explicit internal heat or reservoir variables. A mixed realization may contain both reversible storage and damping into identified internal degrees of freedom. What is not permitted is to use a conservative energy whenever it is convenient and then add untracked damping whenever stability or transport becomes difficult. The selected branch must close its own material, momentum, energy, and, where applicable, entropy ledger.

The bulk consequently functions as an internal reservoir. It may store compression, flow, internal energy, disorder, heat, or other unresolved excitations of the same substance. It is neither an empty vacuum that pulls on the brane nor an inexhaustible battery. Any sustained throat process must alter that reservoir and eventually be solved together with the local brane and throat configuration.

2.3 Light in the brane

Light is proposed to be transverse shear motion guided by the ordered slab. For a wave traveling along one brane direction, its displacement lies within the brane but perpendicular to the direction of travel. In the four-dimensional parent space, excluding the one normal direction and the one propagation direction leaves two independent tangential polarization directions. Under the currently supplied homogeneous, isotropic quadratic brane action, that geometrical count is realized as two transverse modes with characteristic speed

$$c_\gamma^2 = \frac{\mu_R}{\rho_{\text{br}}},$$

where μ_R is an effective shear modulus and ρ_{br} is an effective brane inertia density. The same quadratic system contains one separate in-plane longitudinal mode and no transverse–longitudinal cross-block at that order. This is a conditional result of the stated homogeneous action, not yet a proof of confinement on a finite nonuniform slab.

Three related objects must remain distinct. A *background light wave* is a small-amplitude transverse excitation of the throat-free brane; ordinary plane waves and spreading linear packets belong here. A *free photon candidate* is a much stronger object: a finite-energy traveling packet localized in all three brane directions, guided across w , and stable or sufficiently long-lived. A *throat support mode* is a transverse standing mode bound to or resonant around a particle throat. It may help hold the throat open even if free light does not form a self-bound packet in empty brane space.

Guidance across w and localization within the brane are separate problems. The finite slab may prevent a transverse mode from leaking normally into the bulk while its envelope still spreads in x , y , or along its propagation direction. The stronger photon hypothesis therefore asks whether finite-thickness dispersion and nonlinear material response can maintain a three-dimensional packet.

The intended packet remains predominantly transverse. Its intensity may excite a reflection-even dressing made from some combination of longitudinal compression, thickness change, density change, and order change. If that helper response is determined by the transverse intensity rather than independently selectable, it can alter the local propagation environment without becoming a third photon polarization. Because the dressing is reflection even, ordinary free light can use the slab’s normal structure without acquiring the leading reflection-odd electric monopole assigned to charged throats.

The longitudinal branch is therefore neither charge nor a mode to be discarded merely because exact Maxwell theory would not contain it in the same form. It may contribute to the slaved helper response, remain as an independent guided wave, leak into bulk sound, or become an unwanted radiation and drag channel. The complete finite-slab spectrum must decide. A viable light sector must also control polarization splitting, bulk leakage, defect-induced conversion, dispersion, and any nonlinear wake. Failure of the self-bound packet would leave a possible classical two-polarization light-wave sector, but it would not complete the intended ontology of an individual freely propagating photon.

2.4 Particles as finite supported throats

The ontology contains no fundamental point particles. A particle is proposed to be a finite nonlinear throat, puncture, or defect in which the ordered brane extends into the bulk. A candidate branch may be labeled

$$\mathcal{T}_{\alpha,s}, \quad s \in \{+1, -1\}.$$

The label α contains orientation-independent species information: the support-mode family, core topology, boundary data, equilibrium size and shape, transport and conversion pattern, and any additional stable or metastable internal structure. The sign s records whether the throat is oriented toward $+w$ or $-w$. Reversing s does not by itself turn one species into an unrelated species.

The complete particle is more than its apparent mouth. It includes the full four-dimensional throat geometry; the internal transverse standing mode; the ordered rim, collar, sheath, cavity wall, or other region in which support stress is concentrated; surrounding flow, pressure, density, order, and brane deformation; the conversion region; its long-range fields; and the return and reservoir response required by the selected dynamical branch. Here return is the separate distributed porous-brane S_{leak} process surrounding the object, not reverse flow through its throat. A brane observer may assign the object one projected location, but the physical core occupies a finite region of $(\mathbf{x}, w)$.

A trapped transverse standing mode is proposed to help hold the aperture open. Its stress is structural. The model does not identify the support-mode energy with observed gravitational or inertial mass, and it does not assume a relation such as $E_{\text{support}} = Mc^2$ as an input. The equilibrium throat must instead arise from a balance among support stress, brane tension, material transport, order conversion, bulk backpressure, and the reservoir state.

A linear bound mode on a provisional throat geometry is only a spectral seed. The actual particle must be a self-consistent nonlinear solution in which the oscillating support field and the time-averaged geometry, flow, conversion, and reservoir loading agree. The throat can therefore be stationary in its average shape and rates while its support field remains periodic. Its stability may require a Floquet or retarded-response analysis rather than the Hessian of a strictly static energy.

Material throughput and order conversion are related but distinct parts of the throat. The net constituent flow across an aperture, denoted schematically by Q_n on a selected outward-drain branch, measures transport. A gross local ordered-to-bulk conversion diagnostic, denoted Q_χ , measures loss of resolved brane order in the throat region. They need not be equal, and their precise definitions depend on the chosen reversible, relaxational, or mixed order branch. Neither quantity is a return rate: material that passes through the throat does not come back through that aperture. Return is instead the separate distributed reordering of bulk material into the porous brane, represented by the positive bulk-to-brane contribution to S_{leak} .

Different particle species may correspond to different solved throat branches, not merely to different electric orientations. Species structure could reside in distinct support modes, knot or core topology, composite geometry, or other orientation-independent data. Consequently, reversing an electron-like throat produces the leading positron-like candidate on the same internal branch; it does not produce a proton-like object. A proton-like object would require a different or composite branch.

The particle hypothesis remains open. No solution has yet jointly established a finite supported geometry, localized or sufficiently long-lived support mode, acceptable throughput and conversion, stable source profiles, reservoir loading, and perturbative or Floquet stability. Later throat work is intended to decide whether this picture can be realized by the frozen parent medium.

2.5 Charge, gravity, magnetism, and radiation

2.5.1 Electric orientation and finite-thickness charge

Electric sign is identified with throat orientation:

$$s = +1 \quad \text{for a throat toward } +w, \quad s = -1 \quad \text{for a throat toward } -w.$$

The sign is an intrinsic property of an oriented defect, but it does not by itself determine the magnitude of the long-range coupling, the particle species, or the complete antiparticle operation. For a finite slab, an oriented throat generally perturbs both interfaces. Its leading mid-surface-like disturbance is reflection odd and reverses with s , while a thickness-like disturbance is reflection even and can remain the same for both orientations.

The long-range electric carrier is therefore not assumed to be the bare height of one interface. It must emerge as a healthy odd eigencombination of every relevant coupled variable. To produce a Coulomb-like potential in three brane dimensions, the static long-wavelength operator must contain a branch with the appropriate leading k^2 stiffness. Reflection oddness or gaplessness alone is insufficient; a different low-wave-number stiffness would produce a different range law.

The intended force is a material compatibility effect. A throat imposes an oriented boundary condition on the finite slab. Same-oriented defects should compound the departure from the preferred brane state, while opposite-oriented defects should partly cancel it. The actual force sign must be derived from the correct equilibrium boundary ensemble or, for periodic or driven throats, from cycle-averaged stress, momentum flux, conversion, and reservoir response. It cannot be fixed merely by naming a variable “electric.”

At long range the desired organization is

$$q_a = s_a g_a, \quad g_a \geq 0,$$

where s_a carries all electric sign and g_a is a derived coupling magnitude. Approximate unit-charge universality requires ordinary one-charge throat branches to produce essentially the same leading g_a , even when their internal species structures differ. A common slab thickness helps but does not prove this; the core boundary data must also be universal, quantized, topologically controlled, or sufficiently decoupled from the far field.

A binary orientation label also does not automatically provide additive charge conservation. The model must derive a conserved oriented count or current and show how it appears in a projected Gauss-like law. Likewise, the complete antiparticle candidate may require more than geometric reflection. In general the target operation is

$$\mathcal{C}_{\text{th}} = \mathcal{I}_{\text{internal}} \circ \mathcal{R}_w,$$

where $\mathcal{R}_w$ reverses the embedding orientation and $\mathcal{I}_{\text{internal}}$ transforms any additional odd phase, circulation, chirality, support-mode convention, topological datum, or conversion current. Whether the internal transformation is trivial is an output of each solved species branch.

Finite thickness prevents electrical cancellation from being mistaken for material disappearance. Opposite orientations may occupy the same projected position while their cores remain on opposite sides of the slab. Even if the complete odd electric source cancels, the even thickness deformation and the internal support structures may remain. The required logical separation is

odd electric cancellation $\nrightarrow$ even material cancellation $\nrightarrow$ core contact $\nrightarrow$ annihilation.

A same-species throat–antithroat pair may eventually annihilate, reconnect, scatter, or form a temporary neutral state. Oppositely charged objects of different species may instead retain unmatched internal structure and form a stable neutral composite. Electric neutrality alone does not determine the short-range outcome.

2.5.2 Gravity as a source-side material response

Gravity is proposed to arise from the complete material configuration associated with an open throat, not from the trapped standing-wave energy alone. The phrase “gravity is inflow” is a useful first picture for the localized one-way throat drain, but the relevant source may also include pressure, acceleration of the medium, ordered stress, de-structuring, the separate distributed porous-brane S_{leak} return, bulk compression, interface response, and reservoir backreaction.

The environmental medium must first produce a probe-independent source field. One fixed operational calibration then maps that field into the acceleration units used by a brane observer. Only after that calibration is frozen is the passive response of an arbitrary test throat calculated. A reference object may set units once, but the calibration may not absorb species-dependent response and make universal free fall true by definition. The calibrated gravity observable is therefore not assumed to be numerically identical to the raw four-dimensional material velocity.

Electric orientation and gravitational sign obey different tests. Under the complete reflected throat operation and reflected environmental data, the ordinary gravity source should remain the same when s reverses. Gravity is targeted to be orientation even, while charge is orientation odd. That parity test does not establish attraction: the local radial sign must separately be shown to point inward on the ordinary branch.

The separate distributed return—bulk material reorganizing into the ordered state across the porous brane, represented by the positive bulk-to-brane contribution to S_{leak} —can change the enclosed gravity source with distance. It may leave an intermediate inverse-square regime, screen the source beyond a return scale, overshoot, or even reverse the effective sign. These outcomes must remain visible rather than being removed by an absolute-value convention. The ontology therefore distinguishes a local active source from the scale-dependent effective source after return is included. Earlier effective calculations coupling a localized sink to a separate distributed porous-brane return

support inverse-square scaling and leading attraction under stated assumptions, but the full source reduction, calibration, parity closure, dynamics, and return profile remain open.

2.5.3 Magnetism and radiation from motion

Electromagnetic magnetism is proposed to be the sign-dependent transverse response of the same oriented throat when it moves. At low speed the source has the schematic form

$$\mathbf{J}_T \propto s\mathbf{V}.$$

Reversing the throat orientation or the velocity reverses the magnetic source. The coefficient, sign, tensor structure, and retarded response must ultimately come from the solved moving throat rather than from an independently assigned magnetic rule.

The intended electromagnetic hierarchy is

$$O(v^0) : \text{static electricity}, \quad O(v) : \text{magnetism}, \quad O(a) : \text{transverse radiation}.$$

One conserved oriented current and one shared coupling structure must control all three orders. A static electric coefficient, a separate magnetic normalization, and a third radiation coupling would not establish one medium-based electromagnetic sector.

Several motion-induced effects must remain separate. Electromagnetic magnetism is sign dependent. The inertial wake or rebuilding of the surrounding medium is primarily sign independent. Motion may also alter throat throughput and thereby modify gravity. Gravitomagnetism is vorticity or frame-dragging-like structure in the gravity-associated flow and is not the same phenomenon as electromagnetic magnetism.

Acceleration must be tested for transfer of energy and momentum into both transverse light polarizations. The same event may also emit into the odd electric branch, even thickness modes, the longitudinal brane mode, bulk sound, or order and reservoir excitations. Those additional channels are calculable predictions. They may not be omitted simply because they have no counterpart in exact Maxwell theory.

2.6 Inertia and the complete dressed particle

Inertia is proposed to be the resistance of the complete dressed throat to acceleration. The particle is not a rigid mouth moving through an otherwise unchanged background. It carries an oscillating support mode, a deformed brane, near-field flow, pressure and density disturbances, a conversion region, and a reservoir response. Changing the throat's velocity requires some or all of that configuration to be rebuilt.

The trapped standing mode may need to retime its counterpropagating components, shift phase and nodes, or change its resonant geometry as the throat accelerates. Momentum must be transferred into that reconfiguration even if the local transverse wave speed remains unchanged. The surrounding medium must also adjust its flow, pressure, order, and deformation. Reversible rebuilding can

contribute an added-inertia-like response; outgoing waves and irreversible lag instead appear as radiation or damping and must be accounted for separately.

The word *mass* therefore refers to three distinct observational roles:

Role	Meaning in this proposal
Active gravitational mass	The source strength inferred from the local gravity field after the source-side reduction and one fixed calibration.
Passive gravitational mass	The low-frequency coefficient governing how a test throat responds to an already defined external gravity field.
Inertial mass	The low-frequency coefficient governing resistance of the complete dressed throat to acceleration.

These roles may arise from different pieces of the same solved object. Their equality is not assumed. The ordinary branch must first produce positive active, passive, and inertial behavior under fixed conventions. It must then show that the appropriate low-energy ratios are sufficiently universal across species. Universality of a wrong-sign, singular, or unstable response would not count as success.

The calculation depends on the dynamical branch. In a closed reversible theory, inertia may be extracted from the velocity dependence of the total momentum of the support mode, geometry, near fields, and conversion sector. In a relaxational or mixed theory, the response is frequency dependent and must include damping, memory, internal heat or reservoir variables, and entropy production. An unexcited reservoir may not deliver net work during a small periodic perturbation; the retarded response must be passive.

The preferred material frame also creates a possible failure channel. Uniform motion may emit into any branch whose phase velocity permits it, and a relaxational medium may produce friction even when no propagating wave is radiated. The full spectrum and low-frequency response must therefore be checked for Cherenkov-like thresholds, modes with vanishing critical velocity, conversion lag, and other forms of vacuum drag. The model has identified plausible sources of inertia, but no moving supported-throat family has yet derived a positive inertial response, acceptable drag, or the required universality.

2.7 Global conservation, return, and reservoir closure

A throat provides a localized, one-way drain from the ordered brane. Material passing through the throat mouth de-structures into the bulk state; it does not return through the throat. Net aperture throughput tracks where material moves, while gross local order conversion tracks how much resolved brane order is lost in the throat region. These are not automatically the same rate, and neither is the distributed return rate. A correct branch must derive all three without hiding them inside one adjustable “drain strength.”

Return is a separate, spatially distributed process. The finite-thickness brane is porous, so bulk material can cross into the resolved brane region and reorganize into the ordered brane state away from the throat mouth. For a projected brane observer, constituent continuity has the form

$$\partial_t \rho_{\text{brane}} + \nabla_3 \cdot \mathbf{j}_{\text{brane}} = S_{\text{leak}}.$$

The positive bulk-to-brane contribution to the signed exchange term S_{leak} represents distributed return. The full S_{leak} is not synonymous with return: it is the net transverse exchange seen by the projected observer and may have either sign. In an explicit two-state order ledger, transport and order conversion are refined further, but none of these descriptions permits a local down-and-back-up throat loop.

In a closed realization the material cycle is

ordered brane material $\longrightarrow$ localized one-way throat drainage
 $\longrightarrow$ throat-region de-structuring and bulk reservoir
 $\longrightarrow$ distributed S_{leak} reordering across the porous brane
 $\longrightarrow$ ordered brane material.

The bulk may store density and pressure changes, flow, internal energy, disorder, heat, or other excitations. The return leg is not reverse throat flow. Backpressure or chemical-potential imbalance within the same medium may drive bulk material to reorganize into the ordered state across the porous brane. It may not be supplied by an unchanged external pump.

The local throat drain and the global porous-brane return–reservoir system are nevertheless one feedback problem. A provisional local solution produces one-way throughput, conversion, stress, emitted waves, and energy transfer. Those outputs alter the global bulk, distributed return profile, and reservoir state. The changed reservoir determines the distributed S_{leak} return across the brane and also alters the backpressure and asymptotic data seen by the throat; it does not send material back through the throat. Closure requires iteration until the local throat and global solution reproduce the same boundary conditions, fluxes, distributed return profile, and reservoir state. A local throat is not fully solved if its own long-term operation would invalidate the environmental data used to obtain it.

This feedback also closes the gravity problem. The separate distributed porous-brane return contribution changes the source enclosed at larger radii and may determine whether a useful local inverse-square region survives, becomes screened, overshoots, or changes sign. A force law derived while holding an inconsistent reservoir or return profile fixed is only a provisional local result.

If the selected branch is relaxational or mixed, the global ledger must also track entropy, heat, and degraded energy. A permanent particle cannot consume ordered structure forever while the reservoir and ordered brane remain unchanged. In a reversible branch, the localized one-way throat drain and the separate porous-brane return must instead form parts of a coherent globally closed transport or periodic cycle with complete energy and momentum accounting. Periodicity of the global cycle does not imply reversal of the throat-mouth flow.

A possible later cosmological extension asks whether the spatially integrated distributed porous-brane return rate could slowly exceed the integrated localized throat-drain rate and thereby increase the intrinsic three-volume of the ordered brane. That idea is downstream of local–global closure. Return does not by itself imply expansion, accelerated expansion, or a dark-energy analog; each would require a separate global derivation.

2.8 What is postulated and what remains to be shown

The ontology is intentionally more mature than the derivation status. It defines a coherent object of study and prevents later calculations from changing the meaning of the model, but it does not convert open mechanisms into results. The main status distinctions can be summarized compactly:

Element	Present epistemic status
Parent arena and one conserved medium	Postulated: preferred time, four Euclidean spatial dimensions, one compressible substrate, and global constituent conservation.
Ordered finite brane and bulk state	The two-state ontology and shear-supporting intent are postulated; a dynamically selected stable slab, microscopic shear origin, and complete spectrum remain open.
Light	Two transverse modes and one longitudinal mode follow conditionally from the supplied homogeneous quadratic action; finite-slab guidance, acceptable leakage, and a localized photon packet remain open.
Particles	Finite oriented supported throats are the working particle hypothesis; no jointly self-consistent and stable throat solution has yet closed geometry, support, transport, conversion, and reservoir loading.
Electric sector	Orientation supplies the postulated sign; the Coulomb-carrying branch, interaction sign, universal coupling, conserved current, complete antiparticle map, and short-range dynamics remain to be derived.
Gravity	Localized one-way throat drainage together with separate distributed porous-brane return is the proposed mechanism; inverse-square scaling has conditional precursor support, while the full source reduction, calibration, parity, sign, dynamics, and S_{leak} return profile remain open.
Magnetism and radiation	Moving-source tensor and falloff structure have conditional precursor support; one-current normalization across electricity, magnetism, and all radiation channels remains a target.
Inertia and global closure	Reconfiguration of the complete dressed throat is the proposed origin of inertia; positive response, low-energy universality, acceptable drag, and the local-global reservoir fixed point remain open.

Several claims are deliberately outside this ontology. The model does not claim to be exact general relativity or exact electromagnetism. It does not insert a fundamental electromagnetic gauge field merely to recover Maxwell's equations, and it does not identify the longitudinal brane mode with charge. It does not claim that support-mode energy is observed mass, that active, passive, and inertial mass share one microscopic mechanism, or that all characteristic speeds are automatically equal. It does not yet supply quantum theory, photon number, particle statistics, or a derivation of the real universe from first principles.

The resulting mental picture is nevertheless unified. One conserved four-dimensional medium forms an ordered finite-thickness brane inside a de-structured bulk state. The brane carries transverse light and other collective modes. Finite oriented throats are candidate particles. Orientation supplies

electric sign, while complete throat structure supplies species. Localized one-way throat transport, stress, and conversion, together with the separate distributed porous-brane S_{leak} return and reservoir response, are proposed to underlie gravity. Motion and acceleration produce magnetic, radiative, and inertial effects. The bulk stores the material and energy involved and supplies material for distributed reordering into the porous brane; it does not return material through a throat.

The research program asks whether this entire story can be realized by one stable parent system without independent retuning. Success requires the same fields, coefficients, branch choice, source conventions, and reservoir data to survive every sector. Failure to obtain that common realization would reject the candidate branch, or ultimately the architecture, even if individual analogies remain suggestive.

Research Question and Strategy

The ontology in chapter 2 defines one proposed material picture. This chapter defines how that picture will be tested. The program is not organized to accumulate separate analogies for familiar phenomena. It is organized to determine whether one controlled brane–bulk medium can satisfy the full set of light, force, particle, and reservoir requirements at the same time. That distinction controls the order of work, the use of provisional sources, the treatment of parameters, and the interpretation of both positive and negative results.

3.1 Central research question and scope

The central research question is:

Does there exist at least one mathematically coherent one-medium brane–bulk system whose common field content, constitutive laws, dynamical branch, boundary conditions, and internal reservoir structure can simultaneously support the required far-field sectors, admit stable particle-like throats that source and respond to those sectors consistently, and close the global material cycle without independent sector-by-sector retuning?

This is an existential compatibility test. Let Λ_S denote the region of admissible common-medium choices that satisfies every mandatory condition imposed by sector S . The proposed architecture survives the far-field compatibility stage only if

$$\Lambda_{\text{common}} = \bigcap_{S \in \mathcal{S}_{\text{mandatory}}} \Lambda_S \neq \emptyset.$$

The intersection, rather than the union of separately successful regions, is the research object. A light calculation at one parameter point, an electric calculation at another, and a gravity calculation on a different dynamical branch do not together establish one medium. Nor does the absence of an immediate contradiction establish that the intersection is nonempty. The program must construct and test at least one common realization.

The scope is deliberately bounded. This is a classical toy-model research program. It may postulate an arena, material states, constitutive terms, and a permitted operator basis, then test

the consequences of those choices. Even a fully successful program would establish the existence of a coherent classical analog only within its declared regimes. It would not by itself establish that the real universe is literally constituted this way, reproduce exact general relativity or exact electromagnetism, or supply quantum theory, photon statistics, or a complete particle spectrum.

Six epistemic distinctions are sufficient for the proposal. A *postulate* defines the candidate architecture. An *operational definition* fixes how a brane observer measures or calibrates a quantity. A *target relation* is an output the model must produce, not an answer that may be inserted into the equations. A result *derived under stated assumptions* remains limited to its branch, boundary ensemble, approximation, and regime. An *open hypothesis* identifies an unresolved mechanism or solution. A *failure criterion* states what result rejects a named branch, candidate, mechanism, or architecture. The canonical ontology and closure ledger control those statuses in detail; this proposal uses them only to keep the claims proportionate to the evidence.

3.2 Capability-first, far-field-first reasoning

The constitutive structure of the final parent medium is intentionally not fully frozen at the start. Freezing detailed equations before the required capabilities are understood would risk spending substantial effort on a stable slab or nonlinear throat that cannot support a Coulomb carrier, cannot guide transverse light, or necessarily opens a fatal loss channel. The opposite risk is equally serious: adding a new term whenever a later sector becomes difficult would produce several tailored effective theories rather than one medium.

The program therefore begins by deriving *capability contracts*. Each contract works backward from an observer-facing behavior to the minimum source, response, symmetry, spectral, conservation, and stability properties that any successful common medium must possess. It remains broad enough to compare more than one possible realization, but concrete enough to expose incompatible mode counts, signs, stiffnesses, characteristic speeds, couplings, reservoir costs, or boundary conditions.

The far field is the useful first abstraction because it separates long-range behavior from the unresolved nonlinear core. At distances large compared with a throat radius and, where relevant, the brane thickness, a localized object can initially be represented by provisional source data coupled to the background response operators. This permits calculation of required ranges, source parities, Green-function structure, retarded poles, force signs, shared normalizations, leakage, radiation thresholds, and unwanted modes before a complete throat exists.

The provisional source is a disciplined placeholder, not a solved particle. Its amplitude, form factor, conservation law, universality, and relation to an actual throat remain later tests. A far-field result can therefore establish that a particular source *would require* a healthy odd k^2 -stiffness branch, a specified transverse pole, or an orientation-even gravity response. It cannot establish that the common medium admits a stable throat with the assumed source profile. That gap is closed only after the common background survives and the nonlinear throat program replaces the placeholders with solved objects.

The strategy has three broad phases. First, light and the far-field force sectors are converted into compatible requirement statements. Second, those requirements are intersected and realized, if possible, by one frozen Candidate Parent System whose stable brane and complete spectrum are

solved and from which every far-field sector is rederived. Third, only a surviving candidate advances to self-consistent throats, moving and accelerating solutions, two-body contact, and the local-global reservoir fixed point. Compatibility on paper is not existence, and existence of a background is not existence of a particle.

3.3 Requirement-discovery order and mathematical closure order

Two orders of work must remain distinct. The *requirement-discovery order* is the sequence chosen to learn what the common medium must do. The *mathematical closure order* is the dependency chain that determines what must actually exist before a later claim is valid for one candidate. Calendar scheduling may place conditional calculations in parallel, but it does not alter either logical order.

The proposal uses the following discovery sequence:

1. guided light and the stronger localized photon-packet hypothesis;
2. gravity as the combined response to localized one-way throat drainage and separate distributed porous-brane S_{leak} return;
3. static electricity and oriented charge;
4. magnetism from a moving oriented source;
5. accelerating charge and dynamic electromagnetic closure;
6. passive response, inertia, and low-energy universality;
7. the complete audit of extra modes, speeds, leakage, radiation, and drag;
8. cross-sector synthesis and the common viable-region test.

Light leads because it tests the host slab and its shear-supporting spectrum without requiring a solved particle core. Gravity then establishes the source-versus-probe distinction and introduces the burden of keeping localized one-way throat drainage distinct from distributed porous-brane S_{leak} return and global reservoir response. Static electricity adds the reflection-odd long-range sector; magnetism and radiation test whether that sector belongs to one conserved dynamical source hierarchy. Passive and inertial response then test the complete dressed object, while the full spectral audit collects consequences that may cross several sectors at once.

The mathematical closure order instead follows the actual dependency chain:

parent dynamical framework $\longrightarrow$ stable finite brane
 $\longrightarrow$ complete background spectrum
 $\longrightarrow$ self-consistent supported throat
 $\longrightarrow$ throat stability and internal spectrum
 $\longrightarrow$ moving and accelerating throat family
 $\longrightarrow$ two-throat interactions
 $\longrightarrow$ global material-cycle and reservoir closure.

A later stage cannot confer existence on a missing predecessor. A photon envelope may be explored on a provisional slab, or a far-field pair kernel may be studied with provisional sources, but those results remain conditional until the same profiles and coefficients arise from the frozen background and throat solutions. The final local and global stages form an iteration rather than a one-way dependency chain: localized one-way throat drainage changes the reservoir, while the changed reservoir changes the distributed porous-brane S_{leak} return profile and the asymptotic data under which the throat must be solved again; it does not return material through the throat.

3.4 Compact sector-analysis pattern

Every sector chapter follows one compact research pattern. It should answer:

1. What behavior must a brane observer see, and in what regime?
2. What material mechanism is proposed to produce that behavior?
3. What source and response calculation will test the proposal?
4. What static and dynamic behavior is required from the relevant branch?
5. What unwanted modes, leakage, radiation, drag, or departures must be calculated from the same mechanism?
6. What does the result require or forbid in the common medium?
7. What outcome passes, conditionally survives, reopens the work, or rejects the branch?

This pattern prevents a sector from passing by reproducing only its desired leading asymptotic behavior. A static $1/R$ potential does not by itself close the dynamics of its carrier. A correct transverse mode count does not establish finite-slab confinement or a localized photon. A plausible moving-source tensor does not establish its normalization from the electric sector. Each chapter must therefore distinguish equilibrium static operators from retarded dynamic response, and it must include every coupled branch activated by the proposed source.

The detailed calculation may use a Hessian and an ensemble-appropriate energy, a cycle-averaged stress for a conservative periodic solution, a retarded operator for radiation or damping, or a full

nonlinear time evolution. The method is allowed to vary with the physical problem. What remains common is the handoff: each sector returns a normalized set of required fields, coefficients, speeds, sources, constraints, exclusions, and failure conditions for the cross-sector synthesis in chapter 6.

3.5 Frozen parameters and the proposal–ledger cross-check

The common-medium claim is meaningful only if shared inputs retain the same identity and value across dependent calculations. The governing rule is:

Once a constitutive term, coefficient, normalization, sign convention, dynamical branch, source definition, boundary ensemble, or derived upstream quantity has been fixed for a controlled baseline, it is inherited by every dependent calculation. A substantive change creates a new version and reopens all actual downstream dependencies.

Freezing is not a claim that an input is true, unique, or fundamental. It means only that the project is testing a named baseline rather than allowing each sector to select its own preferred theory. During requirement discovery, the program freezes what a successful medium must do and how the decisive outcome will be judged. During common-medium realization, Candidate Parent System V1 freezes the actual field inventory, conservative, relaxational, or mixed branch, equations and constraints, operator basis, symmetries, parameters, boundary and reservoir data, source conventions, and permitted calibration choices.

A candidate may contain a declared parameter region rather than one numerical point, but each tested point must be evaluated across every applicable sector. A favorable result cannot compensate for a ghost, wrong force sign, failed conservation law, missing mode constraint, nonuniversal coupling, unacceptable loss channel, or unclosed reservoir. Derived outputs also cease to be adjustable inputs: a guided-light speed, electric eigenvector, throat source profile, or reservoir state obtained upstream must be propagated downstream with its stated uncertainty and regime.

The proposal and derivation ledger provide independent views of the same test:

Proposal track	Derivation-ledger track
States the observer-facing question and the capability that must be shown	Starts from declared equations, assumptions, branches, and boundary data
Defines the required calculation, unwanted channels, and pass or failure scope	Performs and records the symbolic, analytic, and numerical calculation
Identifies shared quantities and cross-sector dependencies	Records frozen inputs, outputs, provenance, uncertainty, and reopened descendants
Checks whether the mathematics answered the intended physical question	Checks whether the algebra, implementation, and convergence support the stated claim

Where symbolic verification is feasible, the project uses independently constructed SymPy and Mathematica calculations to compare invariant results rather than merely sharing one generated expression. Where symbolic closure is unavailable, numerical evidence must include branch, boundary,

convergence, and stability checks appropriate to the claim. The detailed artifact and review procedures remain in the derivation ledger and chapter 8; the proposal needs only the principle that no claim is promoted without a current bottom-up evidence path.

3.6 Falsifiability and levels of outcome

The breadth of the ontology does not protect it from failure. Falsifiability comes from requiring one frozen system to carry every consequence, separating static from dynamic questions, calculating unwanted channels, keeping source properties distinct from probe response, and closing the material and reservoir ledgers. Words such as “small,” “long lived,” “approximately universal,” or “acceptable” must resolve, before decisive evaluation, either to an exact structural requirement or to a bounded metric with a declared regime and uncertainty. Mandatory conditions are conjunctive; they are not combined into a weighted score that permits success in one sector to pay for failure in another.

A negative result must also be attached to the correct level. Failure of one algebraic implementation or unconverged solver leaves a claim unresolved. Failure of one approximation rejects that approximation. Failure of a permitted solution branch rejects that branch. A controlled violation of a mandatory gate rejects the named Candidate Parent System. The broader architecture fails only when the mandatory capability intersection is shown to be empty throughout the admitted theory class, a constitutive mechanism essential to the ontology is ruled out throughout its permitted realizations, or closure requires prohibited sector-specific repair.

The principal outcomes are therefore layered:

Outcome	Meaning
Sector result	A named behavior or no-go statement is established within its assumptions. It remains valuable without establishing the complete model.
Partial analog	Several sectors survive, or a weaker claim survives after a stronger one fails; for example, a valid linear light-wave sector may remain even if no self-bound photon packet exists.
Candidate failure	One frozen Candidate Parent System violates a mandatory requirement or cannot close its own background, throat, interaction, or reservoir problem.
Architecture failure	The common mandatory region is demonstrably empty across the admitted architecture, or success requires untracked new mechanisms or independent retuning.
Full program survival	One current parent system passes the common far-field intersection, stable background and spectrum, supported-throat, moving, pair, and global-reservoir stages within the declared regimes.

An unresolved calculation is neither a pass nor a failure. It should identify the missing object: a stable slab, source profile, boundary ensemble, dynamic operator, convergence result, acceptance threshold, or global fixed point. Likewise, a positive result authorizes only the claim actually tested. Even full program survival would establish a controlled classical analog architecture, not empirical identification with nature. Conversely, an empty common region, an unavoidable extra mode, a

failed supported-throat branch, or an unclosed reservoir cycle is a legitimate and informative result because it identifies where the proposed one-medium program ceases to be viable.

Light Research Program

Light is the first detailed sector in the requirement-discovery program because it tests the proposed host medium before a particle throat has been solved. A throat-free ordered slab must already possess the right shear support, guided mode structure, polarization count, characteristic speed, dispersion, and isolation from unwanted branches. Those results can be sought directly from the background brane and its perturbation operator. They therefore expose incompatible constitutive choices earlier and more cheaply than beginning with a nonlinear particle core.

The light program contains two nested claims. The first asks whether the common medium supports ordinary long-wavelength transverse light on the finite slab. The second asks whether the same medium can also support a finite, freely propagating classical packet that could serve as an individual photon analog. Success of the second track requires success of the first, but failure of the second would not erase a valid conditional light-wave sector. The detailed sequence of spectral, asymptotic, envelope, and numerical calculations is maintained in the companion document *Guided Light and the Photon-Soliton Research Program*. This chapter states the scientific contract and the outputs that must be handed to the common-medium synthesis.

4.1 Research goals and proposed material picture

The proposed material picture begins with one finite ordered slab embedded in the de-structured phase of the same medium. The ordered state must support in-plane shear, whereas the surrounding bulk need not support the same resolved transverse elastic branch. Light is identified with the lowest physical transverse branch guided by this finite material structure. It is not a second substance and no fundamental electromagnetic field is inserted solely to carry it.

Let the brane possess a local normal axis represented by a unit vector N^A . The sign of N^A distinguishes the two throat orientations used by the charge sector, but ordinary free light should depend at leading order only on the unsigned axis, through reflection-even structures such as $N^A N^B$. For a wave vector tangent to the brane, a light displacement is required to be both tangent to the slab and transverse to the direction of propagation. In the four-dimensional parent space, excluding the normal direction and the propagation direction leaves two independent tangential polarization directions. The complete guided spectrum must realize that count dynamically; geometrical counting alone is not sufficient.

Under the currently supplied homogeneous, isotropic quadratic brane action, the two transverse modes have

$$\omega_T^2 = \frac{\mu_R}{\rho_{\text{br}}} k^2, \quad c_\gamma^2 = \frac{\mu_R}{\rho_{\text{br}}},$$

where μ_R is the effective ordered-state shear modulus and ρ_{br} is the effective inertia density of the brane mode. The same action contains one in-plane longitudinal branch,

$$\omega_L^2 = \frac{B_{\text{comp}}}{\rho_{\text{br}}} k^2, \quad c_L^2 = \frac{B_{\text{comp}}}{\rho_{\text{br}}}.$$

At homogeneous quadratic order its transverse and longitudinal blocks decouple. This is a useful conditional result, but it does not yet establish a stable finite slab, derive the shear modulus from the parent medium, confine a transverse mode across the normal direction, or control mixing caused by interfaces, defects, nonlinearity, and reservoir variables.

Three uses of the transverse sector must remain distinct throughout the research:

1. *Ordinary light* is a small-amplitude guided transverse excitation of the throat-free slab. Plane waves and ordinary spreading packets belong to this sector.
2. A *free photon-packet candidate* is a finite-energy traveling object localized across w and within all three brane directions, with only two freely selectable asymptotic polarizations.
3. A *throat support mode* is a transverse standing mode bound to, or resonant around, a nonlinear particle throat. It may exist even if the free slab does not support a self-bound photon packet.

The immediate goals are therefore ordered. First, determine whether the throat-free slab possesses a healthy gapless guided transverse branch with two polarizations and acceptable leakage. Second, derive its finite-thickness dispersion and nonlinear couplings from the same common constitutive law. Third, test whether those outputs permit a stable or sufficiently long-lived three-dimensional traveling packet. Finally, record every coefficient, profile, speed, continuum threshold, and unwanted coupling that the light calculation imposes on gravity, electricity, magnetism, radiation, inertia, and the later throat-support problem.

4.2 Track A: ordinary guided transverse light

Track A asks for a controlled effective light-wave sector before making the stronger photon claim. The finite slab must act as a genuine waveguide rather than merely being declared one. A variable-coefficient transverse reduction has the schematic form

$$S_T^{(2)} = \frac{1}{2} \int dt d^3x dw \left[\rho_T(w) |\dot{\mathbf{u}}_T|^2 - \mu_{\parallel}(w) |\nabla_{\parallel} \mathbf{u}_T|^2 - \mu_w(w) |\partial_w \mathbf{u}_T|^2 - \mathbf{u}_T \cdot \mathbf{V}_T(w) \mathbf{u}_T \right],$$

with coefficient profiles derived from the solved slab and the selected shear-supporting constitutive structure. This expression is a diagnostic template rather than a frozen final action. The actual

operator may contain higher gradients, nonlocal response, coupling to density, order, thickness, or longitudinal variables, and branch-appropriate damping with its internal reservoir.

The normal problem must identify a lowest transverse profile $f_0(w; k)$ and its projected dispersion $\omega_0(k)$. On a conservative or inertial branch, the intended light mode is gapless:

$$\lambda_0 = 0, \quad \omega_0(k) \longrightarrow 0 \quad \text{as} \quad k \longrightarrow 0.$$

A fixed normal standing frequency that gives $\omega^2 = \Omega_w^2 + c^2 k^2$ would instead describe a gapped branch and cannot serve as ordinary long-wavelength light. On a dissipative or mixed branch, the corresponding requirement is a brane-coupled physical pole or continuous branch of the full retarded response whose real frequency approaches zero in the required way while its damping, spectral weight, and lifetime remain acceptable.

Calling the bulk “unable to carry shear” is not a proof of confinement. A vanishing bulk shear modulus could yield a bound profile, an evanescent tail, a floppy continuum, or strong conversion into other material motion. The complete normal operator and boundary conditions must determine which occurs. A true bound mode requires a well-defined physical norm, finite energy, and acceptable normal decay. A leaky mode must instead be treated as an outgoing resonance with calculated complex frequency, emitted flux, and lifetime. The calculation must also verify that normalization does not hide a profile whose support spreads arbitrarily far into the bulk. Useful diagnostics include the branch’s separation from the relevant essential spectrum, its normal localization length, the fraction of its norm or energy carried by the ordered slab, its effective three-dimensional inertia under a fixed on-brane amplitude, and the residue with which a brane observable couples to the mode.

The two transverse polarizations must remain healthy and degenerate on the homogeneous isotropic background. The full constrained mode census must also retain the separate longitudinal, normal-interface, order, density, and bulk branches. Track A does not require those branches to disappear. It requires that their spectra, couplings, and leakage be known well enough to show that ordinary transverse propagation is not destabilized or accompanied by unacceptable conversion, birefringence, attenuation, or additional radiation. Near a throat or another inhomogeneity, any reactivated transverse-longitudinal or transverse-bulk mixing becomes a calculable prediction rather than an omitted effect.

Finite thickness supplies a material length scale and generally modifies the in-brane dispersion. After projection, the lowest branch may have

$$\omega_T^2(k) = c_\gamma^2 k^2 + a_4 H_{\text{br}}^2 k^4 + a_6 H_{\text{br}}^4 k^6 + \dots$$

The coefficients $a_4, a_6, \dots$ are outputs of the slab reduction. They must be small enough over the intended ordinary-light regime to preserve the required approximately nondispersive propagation, yet their signs and magnitudes may become important for the photon-packet track. They cannot be selected independently for packet binding after the background dispersion has been computed.

A successful Track A result therefore consists of more than the homogeneous formula for c_γ . It must deliver a stable slab profile; the complete transverse eigenfunction; a branch-appropriate

gapless classification; two polarizations; a declared low-dispersion regime; normal localization or an acceptable resonance lifetime; the surrounding continua and coupling strengths; and the dependence of these outputs on the same parameters that enter the force sectors. Until those calculations are completed, the present light-wave result remains conditional on the assumed quadratic action.

The observer-facing contract should be evaluated over a frequency and wave-number family, not at one specially favorable point. The phase and group velocities must be recorded separately, as must any frequency dependence of the polarization vectors and normal profile. Positive energy and a healthy physical norm are required throughout the declared light regime. The two polarization branches should remain degenerate on an isotropic background, and any splitting introduced by curvature, defects, environmental asymmetry, or nonlinear intensity must be computed rather than hidden in an averaged speed. Ordinary free light must also carry no leading reflection-odd electric monopole on a symmetric background. This requirement does not make the light sector exact Maxwell theory: the medium may retain a separate longitudinal branch and other falsifiable material structure. It requires only that the transverse carrier seen as light be cleanly identifiable and that the additional structure remain stable, sufficiently weakly coupled, and observationally acceptable in the stated regime.

Track A must finally provide a normalized carrier for later source calculations. The same mode norm, energy flux, and polarization basis used to describe free light must be used when an accelerating oriented throat is tested for emission. Otherwise the radiation amplitude could be adjusted independently of the background light sector. This later use does not require a solved throat now, but it does require Track A to export unambiguous normalization and flux conventions rather than only a dispersion curve.

4.3 Track B: localized classical photon-packet candidate

A linear guided mode is not yet an individual photon analog. Linear equations permit plane waves and localized superpositions, but they do not normally select a finite packet width, prevent diffraction in three dimensions, or produce a distinguished unit of energy. Guidance across w solves only the normal-confinement problem. A packet moving along, for example, the z direction may still spread in x , y , and along its envelope even when its normal profile remains bound.

The preferred candidate is therefore an oscillatory transverse carrier inside a localized vector envelope,

$$\mathbf{u}_T(\mathbf{x}, w, t) = \text{Re} \left[(\mathbf{e}_1 A_1(\mathbf{x}, t) + \mathbf{e}_2 A_2(\mathbf{x}, t)) f_0(w) e^{i(k_0 z - \omega_0 t)} \right].$$

The normal profile f_0 is inherited from Track A. The complex envelopes A_1 and A_2 carry the two independent polarization amplitudes. The carrier retains a frequency, wavelength, and phase while the envelope localizes energy, momentum, strain, and stress. Compact support of the displacement itself is unnecessary; the physical requirement is finite, concentrated energy and momentum with no unacceptable secular tail.

The proposed binding mechanism combines finite-thickness dispersion with a reflection-even material dressing excited by the transverse intensity

$$I = |A_1|^2 + |A_2|^2.$$

The helper is not assumed to be a pure longitudinal displacement. In the finite slab it may be a derived mixture of in-plane compression, even thickness change, density response, and order response,

$$\eta_{\text{helper}} = a_L \nabla \cdot \mathbf{u}_L + a_t h_t + a_n \delta n + a_\chi \delta \chi_B + \dots.$$

The coefficients and mode composition must come from the complete background spectrum. At weak amplitude the desired hierarchy is a transverse carrier at first order, an even helper sourced by intensity at second order, and feedback on the carrier at third order. If the helper is functionally determined by I , it is a slaved material dressing rather than a third freely selectable photon polarization. Its even reflection parity also allows the packet to use the slab's thickness and compression response without acquiring the leading odd electric monopole assigned to an oriented throat.

Eliminating the helper through its retarded response may generate a nonlocal, frequency-dependent self-interaction for the two transverse envelopes. The general reduction must retain

$$\eta(\omega, \mathbf{q}) = -\alpha \mathcal{G}_\eta^R(\omega, \mathbf{q}) I(\omega, \mathbf{q}).$$

Only in a controlled quasistatic regime may this reduce schematically to

$$\eta(\mathbf{q}) = -\alpha \mathcal{L}_\eta^{-1}(\mathbf{q}) I(\mathbf{q}),$$

so that the packet creates a moving guide in the material through which its own transverse carrier propagates. A focusing response may oppose diffraction; a defocusing response cannot. A simple local focusing nonlinearity is not enough if its energy is unbounded below and the packet collapses. Saturation, competing nonlinearities, higher-gradient terms, or a finite-range nonlocal helper may be required, but every such stabilizing structure must be part of the shared medium rather than a photon-only repair.

Two reduced routes are useful. A Korteweg-de Vries-like one-dimensional solitary shear pulse is an inexpensive test of whether finite-thickness dispersion and amplitude-dependent propagation can balance at all. It is not the preferred final photon model because it does not establish lateral localization, two-polarization carrier structure, or near amplitude-independent light speed. The principal route is the guided oscillatory carrier with a three-dimensional vector envelope and slaved helper. A one-dimensional success is therefore a precursor; it must survive transverse stability, full three-dimensional continuation, and reconstruction in the parent fields.

The reduced vector system must preserve the symmetries of the parent slab. On a homogeneous isotropic background, no fixed in-brane direction may be introduced to collimate one chosen propagation axis. Any effective guide should align with the packet that creates it. Polarization-dependent nonlinear invariants must be derived explicitly so that linearly, circularly, and elliptically polarized families can be compared. A helper sourced only by total intensity may preserve leading

polarization degeneracy, whereas other allowed invariants may split energies, speeds, widths, or stability. Such splitting is a prediction to be bounded, not a coefficient to be independently tuned for each polarization.

Existence of one isolated numerical lump would also be insufficient. The desired result is a connected traveling family over a declared carrier-frequency range, with calculable relations among amplitude, width, speed, energy, momentum, and helper strength. Those relations reveal whether localization is robust or requires tuning to a measure-zero point near collapse, continuum emission, or a change of spectral branch. They also determine whether ordinary low-intensity light approaches the linear Track A carrier smoothly. A packet branch that has no controlled small-amplitude connection, changes speed strongly with energy, or exists only at a single exceptional frequency may still be an interesting solitary material excitation, but it would not satisfy the intended broad classical photon role.

The interpretation of a packet also depends on the selected dynamical branch. A conservative photon candidate must be a finite-energy traveling periodic solution with derived energy and momentum, zero net outgoing flux for an exact bound branch, and appropriate Floquet stability. A conservative resonance must instead carry a calculated decay rate and lifetime. A dissipative or mixed candidate must identify the internal reservoir, supplied and dissipated power, momentum transfer, entropy production, passivity, and depletion or replenishment within the closed one-medium loop. A freely propagating packet may not draw unrecorded power from an unchanged bulk.

Within the intended classical stage, a viable packet should propagate at or very near c_γ , have finite energy and momentum, possess no independent rest configuration with the same conserved data, and approach a light-like relation such as

$$E_\gamma \simeq c_\gamma |\mathbf{P}_\gamma|$$

in its declared regime. It should remain localized for the required lifetime, retain exactly two free polarizations, and interact weakly or soliton-like with other packets in ordinary vacuum conditions. These conditions do not by themselves derive $E = \hbar\omega$, photon number, quantum statistics, or discrete emission and absorption. Those remain outside the present classical packet test.

4.4 Calculations, unwanted channels, and early failure tests

The light program is organized to reject unsuitable branches before expensive four-dimensional nonlinear simulation. Its practical sequence is:

1. solve a stable or sufficiently metastable finite slab and freeze the branch-appropriate background and reservoir data;
2. derive the complete constrained linear spectrum, including guided transverse, longitudinal, interface, density, order, bulk, and reservoir response;
3. project the lowest transverse branch to obtain c_γ , its normal profile, finite-thickness dispersion, leakage, and observational range;

4. derive the first allowed nonlinear invariants and their projections onto both transverse polarizations and every candidate helper or loss branch;
5. obtain a vector envelope/helper system from a controlled multiple-scale reduction rather than assuming a nonlinear Schrödinger equation;
6. apply energy, scaling, collapse, continuum, radiation, and transverse instability tests before seeking full packet families;
7. continue surviving reduced solutions to the parent fields, then test long-time stability, leakage, and two-packet collisions.

The longitudinal branch is central to the unwanted-channel audit. Its presence is not itself a failure and it may participate in the even helper. What matters is whether a moving packet excites a localized comoving response or a persistent wake. Three speed comparisons must remain distinct: the packet speed v_{packet} relative to the free in-plane longitudinal speed c_L , the placement of that longitudinal branch relative to the bulk-sound speed c_{s0} , and the packet speed relative to the bulk continuum. In the simple homogeneous comparison,

$$k_w^2 = k^2 \left(\frac{c_L^2}{c_{s0}^2} - 1 \right),$$

so $c_L = c_{s0}$ is a grazing threshold for the free longitudinal branch. It is not by itself proof of leakage or confinement. Actual radiation also requires nonzero source projection and interface overlap with an outgoing mode. The complete finite-slab operator may shift or replace this simple threshold.

The nonlinear helper produces a broader test than the free longitudinal branch. A traveling packet sources a DC comoving component, carrier harmonics, sidebands, and possibly additional internal frequencies. Every such component must be compared with every receiving longitudinal, bulk, thickness, density, order, odd-electric, and conversion branch using the retarded operator and the actual source overlap. A DC response may remain localized while a second harmonic radiates. Conversely, a kinematically open continuum may remain dark by symmetry or vanishing overlap. Neither outcome may be assumed.

The most useful early gates are summarized in table 4.1.

These gates deliberately distinguish levels of failure. Absence of a stable host slab or a healthy guided transverse branch rejects the selected ordinary light architecture. Failure of a particular helper mode rejects that binding mechanism, not necessarily all possible common-medium branches. Failure of all permitted shared self-binding mechanisms would leave classical guided light but not the intended localized free-photon ontology. A packet found only after adding a private coefficient or ignoring its force-sector consequences does not count as a success.

Several of the early tests can produce useful negative results without a full packet simulation. A virial or scaling identity may show that all local focusing branches collapse or disperse. The projected helper Green function may have the wrong sign. A required harmonic may intersect an outgoing continuum with nonzero overlap for every admissible speed. The one-dimensional precursor may possess an unavoidable transverse instability. A non-normal retarded operator may make an apparently narrow resonance extremely sensitive to perturbations. Each result should be recorded

Table 4.1: Early gates for the light program.

Gate	Required result	Failure meaning
Stable host slab	Bounded or sufficiently metastable finite ordered slab with positive shear support	The intended light architecture has no viable background.
Guided transverse branch	Branch-appropriate gapless mode, two polarizations, and acceptable normal localization or resonance lifetime	Track A fails for the selected common-medium branch.
Ordinary-light regime	Acceptably small dispersion, attenuation, birefringence, and conversion over the declared range	The mode is not an acceptable ordinary light analog in that regime.
Helper sign and boundedness	Even intensity response focuses when required without an unbounded collapse direction	The proposed self-guiding mechanism fails or requires a different common constitutive branch.
Radiation audit	DC, harmonic, and sideband coupling to all unwanted branches is absent, suppressed, or quantitatively acceptable	The packet produces an unacceptable wake, leakage, drag, or lifetime.
Three-dimensional existence	Finite-energy localized traveling family survives transverse and parent-field continuation	Track B fails even if Track A remains conditionally viable.
Stability and collisions	No growing physical mode and weak or soliton-like ordinary interactions	The candidate is not a persistent free photon analog.
Cross-sector inheritance	Every useful coefficient and profile remains compatible with the force and throat sectors without retuning	A light-only success does not define one common medium.

at the narrowest justified level: it may reject a coefficient sign, a helper eigenmode, a dynamical branch, or the complete packet track. These analytic and reduced-model gates are valuable precisely because they prevent a difficult numerical search from being interpreted as evidence for a solution when the governing structure already forbids one.

4.5 Requirements handed to the common-medium synthesis

The light program must return normalized requirements and calculated outputs, not merely the statement that “the brane carries waves.” The principal handoff is summarized in table 4.2.

Table 4.2: Principal light-sector handoff to the common-medium synthesis.

Category	Required output or restriction
Background material	Stable finite slab profile, selected thickness, positive ordered-state shear support, and explicit conservative, relaxational, or mixed branch with its reservoir closure.
Guided carrier	Lowest physical transverse profile $f_0(w; k)$, two polarization vectors, gapless classification, c_γ , mode norm, brane participation, and leakage or resonance lifetime.
Dispersion and scales	Valid ordinary-light range; coefficients such as $a_4, a_6, \dots$; relevant continuum thresholds; and any relation among H_{br} , localization length, and carrier wavelength.
Complete spectrum	Longitudinal, interface, density, order, bulk, and reservoir branches with their parity, characteristic speeds, damping, source overlaps, and dangerous low-threshold channels.
Nonlinear response	Allowed transverse invariants; helper eigenvector and parity; focusing or defocusing sign; saturation, nonlocality, memory, and boundedness; and the same coefficients’ effects on the force sectors.
Radiative closure	Derived power or leakage for packet motion, together with the normalized outgoing modes, projectors, and loss channels needed to evaluate later accelerating-throat emission into both transverse polarizations and every odd, even, longitudinal, bulk, and conversion branch.
Packet outcome	Either a stable or sufficiently long-lived three-dimensional family with its energy, momentum, speed, polarization, lifetime, and collision behavior, or a bounded negative result identifying which requirement cannot be met.
Frozen inheritance	No photon-specific terms; every profile, coefficient, speed, branch choice, and reservoir datum is exported to Candidate Parent System V1 and reopened if an upstream choice changes.

Several cross-sector burdens follow immediately. The same slab thickness and constitutive coefficients that guide light also determine the normal parity spectrum used by electricity, the even thickness response used in finite-size forces, the longitudinal and bulk modes that control drag and extra radiation, and the transverse operator later used by throat support and electromagnetic emission. A gapped even helper may be useful for packet localization and for screening unwanted thickness forces, but its speed, range, and nonlinear couplings must satisfy both roles. Finite-thickness dispersion

useful for binding cannot produce unacceptable vacuum dispersion. A light branch close to a longitudinal or bulk continuum may constrain the characteristic-speed region available to gravity and the electric sector. Most decisively, the same transverse coupling structure must later accept energy and momentum from an accelerating oriented throat in both polarizations without opening excessive competing channels.

The common-medium synthesis in chapter 6 will therefore not score light with a single pass or fail label. It will intersect a light requirement region containing the stable-slab, guided-mode, ordinary-propagation, helper, packet, leakage, and branch-closure conditions with the corresponding regions from the far-field force program. A robust common region would justify freezing those light outputs in Candidate Parent System V1 and rederiving them from its complete constrained spectrum. A fragile region near continuum, instability, or collapse thresholds would remain a risk even if one numerical packet were found. An empty intersection would reject the candidate combination rather than invite sector-specific retuning.

Light thus performs two roles in the proposal. Track A determines whether the medium can carry ordinary two-polarization guided radiation at all. Track B asks whether the same material can turn that radiation into a localized classical photon analog without sacrificing ordinary vacuum behavior. Together they provide an early, unusually direct test of the common background and a precise set of inherited quantities for the force, throat, and reservoir work that follows.

Far-Field Force Program

The far-field program asks what one common brane–bulk medium must do before the project commits to a particular nonlinear particle core. At distances large compared with a throat radius, and where appropriate large compared with the brane thickness, a localized object can be represented provisionally by its symmetry, multipole content, source parity, motion, and unresolved form factors. The background response can then be used to determine the range, sign, propagation, leakage, and unwanted channels that any eventual throat must produce. This does not establish that a stable throat with the assumed source exists. It establishes a capability contract that the later throat solution must inherit.

The sectors are ordered so that each removes freedoms from the next. Gravity first establishes a probe-independent source field, a fixed calibration, and the role of distributed porous-brane return. Static electricity adds the reflection-odd finite-slab response and the requirements for charge conservation and universality. Magnetism asks whether motion of that same oriented source produces the required transverse response without a second normalization. Acceleration then tests whether static electricity, magnetism, and transverse light are three limits of one current and one coupling structure. Passive response, inertia, extra modes, and drag complete the far-field contract by stating what a future dressed throat must do when it is placed in, moved through, and accelerated within the same medium.

5.1 Common method for the far-field sectors

The far-field calculation begins on a declared throat-free slab background with all fields needed by the selected conservative, relaxational, or mixed branch. Let

$$\Phi = (\mathbf{u}_T, u_L, h_m, h_t, \delta n, \delta\theta, \delta\chi_B, \dots)^T$$

stand schematically for the complete constrained perturbation vector. The list is not a commitment that every displayed coordinate remains independent after constraints are solved. It is a reminder that the desired carrier cannot be isolated from other variables merely by naming it. Static equilibrium response, when a genuine equilibrium reduction exists, is governed by a Hessian and static Green matrix,

$$\mathcal{H}(\mathbf{k}), \quad \mathcal{G}_{\text{stat}}(\mathbf{k}) = \mathcal{H}^{-1}(\mathbf{k}),$$

on the physical constrained subspace. Dynamic response is governed separately by a retarded operator and its inverse,

$$\mathcal{O}_R(\omega, \mathbf{k}), \quad \mathcal{G}_R(\omega, \mathbf{k}) = \mathcal{O}_R^{-1}(\omega, \mathbf{k}).$$

The static inverse determines long-range kernels only in an ensemble for which an equilibrium interaction functional is legitimate. The retarded inverse determines poles, phase lag, damping, outgoing radiation, and memory. A conservative but intrinsically periodic source may require a cycle-averaged Hamiltonian, Routhian, averaged action, or Noether-stress calculation rather than a static free energy. A maintained dissipative source requires stress, momentum-flux, conversion, and reservoir accounting and may not be assigned a pair potential unless conservativity is independently established.

Before throats are solved, a localized source is represented by a vector $\mathbf{J}_a$ with its overall amplitude, internal profile, and finite-thickness form factor left symbolic. The placeholder may carry known transformation properties under reflection and motion, but it may not assume that its magnitude is universal, that it is topologically conserved, or that its static and moving normalizations agree. Those are later outputs. For an oriented source it is useful to write

$$\mathbf{J}_{\alpha,s} = s \mathbf{J}_{\alpha}^{\text{odd}} + \mathbf{J}_{\alpha}^{\text{even}}, \quad s \in \{+1, -1\},$$

where the odd and even vectors may each contain several coupled fields. This parity decomposition does not determine range or force sign. It only states how the source transforms on a reflection-symmetric background.

Each sector must perform both a desired-channel calculation and a complete response audit. The desired calculation asks whether the common medium contains the required low-wave-number stiffness, tensor structure, pole, source overlap, and normalization. The audit asks what else the same source excites: additional odd or even branches, the in-plane longitudinal mode, bulk sound, interface and order modes, conversion and reservoir response, or the transverse light continuum. A sector does not pass merely because its leading asymptotic term has the right power law while an omitted branch is unstable, long ranged, strongly radiative, or dissipative.

The expansion must also state its spatial regime. The strict leading far field has

$$R \gg \max(R_{\text{throat}}, H_{\text{br}}),$$

so the source is seen primarily through its lowest surviving projected multipoles. An intermediate regime may satisfy $R \gg R_{\text{throat}}$ while $R \sim H_{\text{br}}$, in which case finite-thickness form factors and heavier normal branches can modify the kernel. The leading contract should identify the asymptotic coefficient without pretending that it controls contact or annihilation, while also recording the scale and operator data needed to compute the crossover later. A result obtained only by replacing the two interfaces with an infinitesimal sheet must be reopened if the neglected thickness is what selects

parity, screening, source normalization, or the sign of a correction.

Normalization must likewise be carried consistently from source to response. For each desired carrier the calculation should specify the physical mode norm, source overlap, observable convention, and flux or stress used to extract a force or emitted power. This prevents the same eigenfunction from being rescaled one way in the static sector and another way in the moving or radiative sector. Dimensional agreement and the desired power law are necessary checks, but they do not replace a common normalization tied to the parent variables and boundary data.

The distinction between capability and realization remains strict. During this chapter, a result may show that a source with a specified parity and continuity law would generate an acceptable far field. After Candidate Parent System V1 is frozen, the stable slab and complete spectrum must reproduce the same operators and coefficients. After a throat is solved, its actual source vector, form factor, throughput, conservation law, and motion must reproduce the provisional inputs. A failure at either handoff reopens or rejects the corresponding claim.

Every sector therefore exports the same kinds of information to chapter 6: the fields and constraints it uses; the static and retarded operator blocks it requires; coefficient signs and allowed ranges; characteristic speeds and screening scales; source parity and conservation requirements; normalization relations; prohibited extra channels; and the regime in which the result is meant to hold. These outputs are conjunctive. A correct range cannot compensate for a wrong sign, and a correct static sign cannot compensate for unacceptable dynamics.

5.2 Gravity

5.2.1 Behavior and proposed source-side material response

The observer-facing target is an approximately central, attractive, inverse-square gravity-like field over a declared local or intermediate regime. The source should be independent of the species later used to probe it, and its leading ordinary behavior should be unchanged when an otherwise identical throat is reflected from the $+w$ orientation to the $-w$ orientation in correspondingly reflected environmental data. At larger distances the field may be modified by distributed porous-brane return, but screening, overshoot, or sign reversal must be calculated explicitly rather than removed by convention.

The proposed material mechanism is broader than the slogan “gravity is inflow.” A supported open throat supplies a localized, one-way constituent drain. Its local region may generate pressure gradients, material acceleration, ordered stress, ordered-to-bulk conversion, bulk compression, and interface deformation. Material that crosses the throat mouth de-structures into the bulk and does not return through that throat. Return is a separate, spatially distributed process in which bulk material reorganizes into brane state across the porous brane; in projected continuity it is represented by the positive bulk-to-brane contribution to S_{leak} . Reservoir backreaction couples the local drain and distributed return without making them one route or one rate. The gravity calculation asks whether a definite combination of these source-side and environmental quantities produces a three-dimensional field before any test throat is introduced. Schematically,

$$\vec{\mathfrak{g}}_{\text{src}} = \mathcal{G}_{\text{src}}[\mathbf{J}_n, \mathbf{v}_{\text{med}}, D_t \mathbf{v}_{\text{med}}, n, P, \chi_B, \boldsymbol{\tau}_{\text{order}}, \Gamma_B, \mathbf{J}_\chi, \mathcal{R}_{\text{porous}}; \mathcal{B}_\infty].$$

Here $\mathcal{R}_{\text{porous}}$ denotes the separate distributed porous-brane S_{leak} return profile and its coupled reservoir state. This equation names candidate inputs; it does not assume that each survives the final reduction or that the map is already known. In particular, the source is not defined by how one selected probe accelerates. A reference convention may fix the observer's acceleration units once,

$$\mathbf{g}_{\text{eff}} = \mathcal{C}_{\text{ref}} \vec{\mathfrak{g}}_{\text{src}},$$

but $\mathcal{C}_{\text{ref}}$ is then frozen. It may not depend on the later test species or be adjusted with distance to hide a changing source. Raw material velocity and the calibrated gravity observable are therefore not identified by notation alone.

For a nearly spherical source, the desired local parameterization is

$$\mathbf{g}_{\text{eff}}(r) = -\frac{C_g \overline{Q}_g(r)}{r^2} \hat{\mathbf{r}},$$

where $\overline{Q}_g(r)$ is an orientation-even but radially signed enclosed source coefficient after the source and environmental reflection test has closed. The target covariance is

$$\mathbf{g}_{\text{eff}}[\mathcal{C}_{\text{th}} \mathcal{T}; \mathcal{R}_w \mathcal{B}_\infty] = \mathbf{g}_{\text{eff}}[\mathcal{T}; \mathcal{B}_\infty].$$

This is not the same as requiring equal coordinate-normal constituent flux. Reflected outward-drain throats may have opposite global- w flux signs while producing the same gravity source. Nor does orientation evenness establish attraction. On the ordinary local branch the independent sign test is

$$\overline{Q}_g(r_{\text{local}}) > 0,$$

in the displayed radial convention.

The separate, spatially distributed bulk-to-brane contribution to S_{leak} makes the gravity source potentially scale dependent. The local one-way throat drain and its ordered-to-bulk conversion region may produce a positive enclosed source, while return distributed across the porous brane over a much larger region reduces it. A useful regime would contain a matching window

$$R_{\text{throat}} \ll r \ll L_{\text{return}},$$

or another scale-independent near-to-far prescription, in which the local coefficient is approximately constant. Beyond the return scale, $\overline{Q}_g(r)$ may approach zero, settle to another value, overshoot, or change sign. The proposal does not select among those outcomes in advance. It requires the global material and reservoir solution to determine them and to show that any resulting departure is compatible with the intended application of the analog.

5.2.2 Required calculations and medium requirements

The first calculation is a source-side reduction on a provisional localized drain and conversion profile. It must begin from exact four-dimensional constituent conservation and the selected order dynamics, project the complete stress and transport response onto the brane, and identify the three-dimensional quantity that has acceleration dimensions after one fixed calibration. Earlier effective calculations coupling a localized sink to a separate distributed porous-brane return motivate an inverse-square branch under stated assumptions, but the current program must determine which combination of transport, pressure, stress, ordered-to-bulk conversion, and the positive bulk-to-brane contribution to S_{leak} supplies the coefficient. It must also keep net throat throughput, gross local order conversion, integrated distributed return, and the final gravity-source coefficient distinct; none may be substituted for another by definition.

The second calculation is the reflected-source comparison. Both orientations of the provisional source must be embedded in reflected environmental data, and the complete parent fields must be transformed. If the ordinary vacuum is itself reflection symmetric, the two source fields should agree in that same environment. If the upper and lower bulk-facing reservoirs differ, the calculation should expose the resulting environmental splitting. A charge- correlated gravity difference may be an unacceptable prediction, but it must not be misidentified as an intrinsic failure of the throat reflection map without first transforming the environment consistently.

The third calculation is the radial matching problem for the local one-way drain and the separate distributed porous-brane S_{leak} return. A local source profile and a distributed return profile must be solved together far enough to obtain $\bar{Q}_g(r)$, not merely its near-source normalization. The calculation must determine whether an inverse-square interval exists, how its coefficient is extracted without an arbitrary radius choice, and where return begins to screen or reshape it. The return profile must be universal, environmentally calculable, or retained as an explicit departure. It cannot be chosen separately for each throat species after the local source has been computed.

The local-drain and distributed-return calculations must also close their material and mechanical ledgers. A stationary source cannot remove ordered material, momentum, or energy from the local slab while leaving the bulk and porous-brane return system unchanged. For a conservative branch, the relevant fluxes and stored energies must close within the full medium. For a relaxational or mixed branch, the internal reservoir, dissipation, and entropy production must be included. This does not require the full cosmological solution during the far-field requirement pass, but it does require enough accounting to rule out a gravity source maintained by an implicit external pump.

The fourth calculation is dynamic. A time-dependent source must be coupled to the full retarded background operator. Schematically,

$$g_i(\omega, \mathbf{k}) = \mathcal{R}_{ij}^g(\omega, \mathbf{k}) \mathcal{S}_j(\omega, \mathbf{k}).$$

The poles of $\mathcal{R}^g$ determine propagation, delay, phase lag, damping, and radiation. The leading local compressional response may be associated with a bulk-sound speed c_s , but the complete gravity response may also involve order, interface, return, and reservoir modes. The proposal therefore does not assign gravity one propagation speed before the coupled operator is solved. Every pole with nonzero source overlap must be recorded, including slow relaxational tails and any vorticity or

frame-dragging-like component of the gravity flow.

These calculations impose several requirements on the common medium. It must support a well-posed source-side reduction that is independent of later probe species; a reflection-covariant ordinary source branch; a positive local radial sign; an inverse-square matching regime or another controlled extraction of local source strength; a calculable return profile; a stable dynamic response with acceptable characteristic speeds and damping; and one frozen calibration. The required source variables must also coexist with the light and electric sectors. A constitutive choice that produces the desired gravity response while destabilizing the slab, opening an unacceptable light continuum, or giving the electric odd branch the wrong stiffness is not a gravity success for the common program.

5.2.3 Failure conditions and handoff

The selected gravity branch fails if no probe-independent environmental field can be constructed; if attraction appears only after inserting the response of a chosen test object into the source definition; if reflected ordinary sources do not yield the same gravity behavior under reflected symmetric data; if the local radial sign is wrong; or if the inverse-square coefficient cannot be extracted without an arbitrary observation radius. It also fails if distributed return produces unacceptable species dependence, screening, overshoot, or sign change; if the response contains ghosts, instabilities, or ill-posed modes within the declared analog; if damping or delay is unacceptable; or if one desired result requires a gravity-only coefficient not propagated through the shared medium.

A narrower negative result need not reject the entire architecture. It may reject one source functional, one order-dynamics branch, one return ansatz, or one candidate calibration. The handoff to Chapter 6 must state the surviving source variables, parity and sign requirements, matching and return scales, retarded poles, characteristic speeds, calibration convention, and every coefficient shared with the slab, electric, light, and reservoir sectors.

5.3 Static electricity and charge

5.3.1 Oriented source, long-range carrier, and interaction sign

The observer-facing static target is a signed long-range interaction whose leading potential scales as $1/R$ and whose leading force scales as $1/R^2$, with like orientations repelling and opposite orientations attracting. The source sign is not assigned as a free scalar charge. It is postulated to arise from the $+w/-w$ orientation of a finite throat and is represented in the far field as an oriented material boundary condition on the two-interface slab.

For the graph-regime interfaces,

$$h_m = \frac{h_+ + h_-}{2}, \quad h_t = \frac{h_+ - h_-}{2},$$

where h_m is reflection odd and h_t is reflection even. These are leading collective coordinates, not guaranteed eigenmodes. The complete odd and even response blocks may mix them with order, density, phase, longitudinal, bulk, and reservoir variables. The static calculation must therefore

diagonalize the complete constrained Hessian rather than assume that one bare height coordinate carries electricity.

A Coulomb-like equilibrium tail in three brane dimensions requires a healthy odd eigenbranch with nonzero source overlap and leading k^2 static stiffness,

$$\lambda_E(k) = \kappa_E k^2 + O(k^4), \quad \kappa_E > 0.$$

Its static Green function then has the target behavior

$$G_E(0, k) \sim \frac{1}{\kappa_E k^2}, \quad G_E(R) \sim \frac{1}{4\pi\kappa_E R}.$$

Reflection oddness is not enough. A gapped odd branch would be screened, while a flexural branch with k^4 static stiffness would have a different spatial kernel even though it may also be gapless dynamically. Conversely, reflection evenness does not guarantee short range. Every even branch with nonzero throat coupling must be classified, and any intended thickness-like screened branch must acquire its gap after all mixing and constraints are included.

The force sign must come from the physical source and boundary ensemble. A healthy scalar field with an ordinary linear source term does not automatically produce the required like-repels behavior. The model instead treats orientation as a material boundary condition: same-oriented disturbances should compound the departure from the preferred slab state, while opposite-oriented disturbances should partially cancel it. To turn that picture into a result, an equilibrium branch must compute the correct constrained or Legendre-transformed on-shell energy with one-body self terms removed. A conservative periodic pair must use a derived cycle-averaged functional or averaged stress. A driven branch must obtain the force from the complete stress, momentum-flux, conversion, and reservoir ledger.

It is useful to isolate the reflection-odd Coulomb monopole coefficient,

$$C_{E,ab}(R) = g_a g_b + \delta C_{ab}^{\text{odd}}(R), \quad g_a, g_b \geq 0,$$

with all electric sign carried by $s_a s_b$. The quantity $C_{E,ab}$ must not absorb even thickness forces, parity-mixed environmental terms, or nonlinear core interactions. In an equilibrium description, the leading odd contribution may be organized as

$$E_{ab}^{\text{odd}}(R) = s_a s_b \frac{g_a g_b + \delta C_{ab}^{\text{odd}}(R)}{R}.$$

At finite distance a positive coefficient is not by itself enough, because the radial derivative of the correction also enters the force. The asymptotic like-repels/opposite-attracts result therefore requires both the correction and its scaled derivative to be subleading over the declared Coulomb regime. The finite-thickness crossover and nonlinear core force remain later calculations, developed in detail in the focused opposite-orientation companion rather than inside this far-field chapter.

The static calculation must also inspect every additional long-range branch. A second odd mode could alter charge universality or introduce an extra force. An even gapless mode could survive

after electric monopole cancellation and produce a species-dependent neutral interaction. A parity-mixed term may appear if the background or reservoirs break reflection symmetry. These are not automatically fatal, but their range, sign, source overlap, and observational role must be known before the electric sector can be handed to the common-medium synthesis.

5.3.2 Universality, conservation, and dynamic status

The long-range coefficient is not complete until its source dependence is understood. If several independently solved unit-charge species later produce an asymptotic coefficient matrix, removal of $s_a s_b$ should leave a leading positive-semidefinite, approximately rank-one structure. That is the test that allows one to assign a nonnegative one-throat magnitude g_a . For ordinary unit-charge branches, the further target is

$$g_a \simeq g_0$$

within the declared regime. Common slab thickness alone does not establish this. The throat boundary amplitude, topology, support state, and normal source profile must either be universal, quantized, topologically controlled, or sufficiently decoupled from the leading electric eigenmode. Species-dependent finite-size and polarization corrections may remain, but they must be subleading and calculable.

The orientation label also requires a conservation structure. On a globally two-sided, co-orientable brane, a candidate additive count is

$$N_s = \sum_a s_a,$$

or, more fundamentally, an oriented intersection number. A successful electric sector must derive why continuous evolution conserves the relevant count, how opposite orientations can be created or removed in pairs, and how the count appears in a projected Gauss-like law. At the effective level the target is a continuity equation

$$\partial_t \rho_s + \nabla \cdot \mathbf{J}_s = 0.$$

This law is not established by writing it. It must descend from the permitted parent evolution and remain valid through motion, deformation, pair creation or annihilation, and the finite-thickness core description. If no global normal or equivalent topological construction exists, the charge definition must be reformulated rather than used unchanged.

The present architecture does not yet possess a demonstrated first-class $U(1)$ Gauss constraint. The intended result is therefore a material and topological charge law whose projected far field is Maxwell-like in the stated regime, not an assumption that exact electromagnetic gauge structure is already present. The derivation must show which constraint or conservation identity replaces the missing structure and what additional longitudinal, normal, or material degrees of freedom remain observable. An additive count that is conserved only in the static ansatz but fails during motion or topology change would not close the charge sector.

The electric carrier's dynamical status is a separate gate. The static Hessian may supply a k^2 inverse without determining whether the corresponding degree of freedom is propagating, constrained, relaxational, diffusive, or part of a mixed response. The complete retarded odd block $\mathcal{O}_{R,\text{odd}}(\omega, \mathbf{k})$ must identify its poles, damping, phase lag, continuum placement, and energy or passivity properties. With ordinary local inertia, the Coulomb branch may have $\omega_E \simeq c_E k$, but this is not assumed before diagonalization. A constrained branch must show how a time-dependent conserved source is satisfied without an omitted propagating degree of freedom. A relaxational branch must close its internal reservoir and show that low-frequency damping, noise or memory, and entropy production are acceptable.

This dynamic classification is important even before the radiation chapter. A propagating electric mode may create an additional radiation channel. A slow mode may produce Cherenkov-like emission or drag for ordinary motion. A diffusive mode may generate persistent lag or attenuation. A mode placed near a bulk or longitudinal continuum may leak. Conversely, a constrained response may reduce radiation but impose a nontrivial instantaneous or elliptic closure that must remain mathematically and materially consistent with the parent system. Static success cannot postpone these questions indefinitely.

The electric handoff therefore includes more than κ_E . It must record the complete odd and even eigenvectors, source form factors, screening lengths, continuum thresholds, leading factorization structure, allowed corrections, conservation law, dynamic branch, characteristic speed or relaxation scale, and all shared coefficients. The same g_a , source profile, and current must then be inherited by magnetism and radiation.

5.3.3 Failure conditions and handoff

The selected electric branch fails if the complete constrained system contains no healthy odd eigenmode with nonzero source overlap and leading k^2 static stiffness; if the desired force sign appears only after choosing an unjustified energy ensemble or inserting a sign by hand; if an additional odd or even long-range mode produces unacceptable forces; or if the full static operator is unbounded, ghostlike, or unstable. It also fails if the leading coupling does not factorize sufficiently to define individual charge magnitudes, if ordinary unit-charge species are unacceptably nonuniversal, or if the asymptotic corrections and their derivatives are not genuinely subleading.

The charge program further fails if no conserved oriented count or current can be defined, if charge depends on an inconsistent global orientation, if the dynamic carrier has unacceptable poles, damping, leakage, or radiation, or if the required result can be obtained only by assigning electricity a private field or coefficient that is absent from light, magnetism, the throat, and the common parent action. Failure of one candidate odd branch may leave another permitted realization open; failure of every admitted route to a conserved, healthy Coulomb carrier would reject the electric architecture of the model.

5.4 Magnetism

5.4.1 Moving oriented source and transverse response

Magnetism is the first direct test that static charge belongs to a dynamical source rather than to an isolated boundary-value analogy. At low speed, motion of the same oriented object should produce a sign-dependent transverse source of schematic form

$$\mathbf{J}_s \simeq \rho_s \mathbf{V}, \quad \mathbf{J}_T \propto s \mathbf{V}.$$

The target brane response is rotationally covariant and transverse, with the leading tensor and distance dependence appropriate to a slowly moving localized charge. In the simplest isotropic far field this means a magnetic-like response whose direction is organized by the cross product of source velocity and separation and whose leading magnitude has the corresponding inverse-square field falloff. The exact coefficient and any medium-dependent corrections must be derived from the shared retarded transverse operator rather than copied from Maxwell theory.

The calculation should begin from the conserved oriented source obtained in the static-electric package, expand a translated source profile to first order in $\mathbf{V}$, and project it onto every transverse background eigenmode. The physical magnetic observable $\mathbf{B}_{\text{eff}}$ must be defined through a fixed source and response convention, including its transformation under orientation reversal, velocity reversal, spatial rotations, and time reversal. The far-zone tensor, power law, phase, and finite-thickness form factor can then be extracted from the retarded Green matrix. Existing exploratory work supports the expected moving-source tensor structure, falloff, and velocity order under stated assumptions, but the source normalization and sign remain conditional on the unresolved electric boundary data and throat interior.

A moving source also perturbs the background in sign-independent ways. The complete calculation must separate at least three effects:

1. electromagnetic magnetism, which reverses with s and with $\mathbf{V}$;
2. inertial rebuilding, compression, and wake structure of the dressed medium, which is not principally electric-sign dependent; and
3. velocity-dependent gravity, including any change in localized throat throughput or conversion, in the separate distributed S_{leak} return profile, or in vorticity of the gravity-associated flow.

The last effect may include a gravitomagnetic or frame-dragging-like component, but it is not electromagnetic magnetism merely because both involve transverse or vortical response. Their source parity, calibration, and coupling to probes must remain distinct.

5.4.2 Shared normalization and unwanted responses

The decisive magnetic requirement is inheritance. If the electric far field has a leading source magnitude g_a , then the same g_a , the same orientation sign s_a , the same normal source form factor, and the same continuity law must determine the leading moving current. The magnetic coefficient may depend on already-frozen transverse-mode normalization and characteristic speed, but it may not be independently fitted. Schematically,

electric source data + source velocity + shared transverse response $\longrightarrow \mathbf{B}_{\text{eff}}$.

This relation is stronger than obtaining the correct tensor form. Many media can produce a velocity-dependent transverse field. The unification claim requires the electric and magnetic normalizations, relative sign, and source dependence to arise from one coupling structure. A source whose static response couples to one eigenmode while its moving response requires an unrelated coefficient or a different current has not closed the electromagnetic sector.

The unwanted-response audit must include excitation of the odd electric mode, even thickness and order modes, the longitudinal branch, bulk sound, transverse light, and conversion or reservoir modes. Uniform slow motion should not continuously radiate below the applicable thresholds. Any stationary comoving wake must be distinguished from an outgoing wave, and any dissipative phase lag must be included in the force and power balance. The calculation must also test whether the magnetic observable acquires unacceptable dependence on the particle species, internal support state, direction relative to a preferred medium frame, slab asymmetry, or environmental reservoir.

Because the target is Maxwell-like rather than exact Maxwell theory, the calculation must also record any time-reversal or constitutive departure carried by the magnetic observable. Such a departure is not automatically fatal in a toy analog, but it must remain compatible with the static source law, the radiation response, and the declared observer-facing regime. Calling a transverse material swirl $\mathbf{B}_{\text{eff}}$ does not by itself establish its transformation properties or its force on a later moving probe.

The magnetic branch fails if it has the wrong orientation or velocity parity, wrong leading tensor or falloff, instability, or unacceptable extra channels. It also fails as an electromagnetic-unification result if its normalization can be obtained only by a new independent coupling. Its handoff to Chapter 6 is the complete first-order moving-source kernel, tied normalization, transformation properties, characteristic speed and phase behavior, and a separated inventory of magnetic, inertial, and gravity-associated responses.

5.5 Accelerating charge and electromagnetic radiation

5.5.1 One current across electricity, magnetism, and light

Acceleration tests whether the static, moving, and radiative limits belong to one source hierarchy,

$$O(v^0) : \text{static electricity}, \quad O(v) : \text{magnetism}, \quad O(a) : \text{transverse radiation}.$$

The target is not simply that an accelerating defect makes some waves. The same conserved oriented density and current, with the same leading one-throat coupling magnitude and source profile, must generate all three orders. The normalized transverse light modes obtained in chapter 4 provide the receiving states. The accelerating source calculation must show how energy and momentum leave the dressed source and enter both physical transverse polarizations with no separately chosen radiation coefficient.

At the effective level the source obeys

$$\partial_t \rho_s + \nabla \cdot \mathbf{J}_s = 0,$$

while its acceleration-dependent perturbation is projected through the complete retarded response. For each normalized outgoing light mode $\lambda = 1, 2$, the far-zone amplitude is determined by the overlap of the source with that mode's eigenvector, normal profile, and flux normalization. The exact formula is candidate dependent, but the logic is fixed:

$$(\rho_s, \mathbf{J}_s; g_a) \xrightarrow{\mathcal{G}_R} \{\mathcal{A}_1, \mathcal{A}_2, \mathcal{A}_{\text{extra}}\}.$$

The emitted transverse power, polarization pattern, angular dependence, and frequency scaling must follow from those amplitudes and the same energy-momentum normalization used for free light. A calculation that rescales the outgoing modes or source separately from the light chapter would not test a shared coupling.

The source-side energy balance must include reversible storage as well as radiation. Acceleration can reconfigure the standing support mode, throat geometry, near-field electric and gravity response, pressure distribution, and conversion region. The external power should therefore be organized schematically as

$$P_{\text{ext}} = \frac{dE_{\text{dressed}}}{dt} + P_{u_T} + P_{\text{extra}} + P_{\text{diss}},$$

with momentum carried by the corresponding fields and reservoir. Reversible near-field rebuilding contributes to the time derivative and later to inertia; it must not be counted as radiation. Conversely, outgoing flux may not be hidden inside a frequency-dependent inertial coefficient without being included in the power ledger.

This stage also fixes the dynamic interpretation of the electric carrier. If it is a propagating mode, its acceleration-induced radiation must be calculated. If it is constrained, the constraint must remain compatible with the conserved source and the retarded transverse response. If it is relaxational or mixed, its phase lag, dissipation, internal reservoir, and possible stochastic partner must be explicit. Static Coulomb behavior, magnetic response, and transverse light emission cannot be declared unified while the main electric degree of freedom remains dynamically unspecified.

5.5.2 Additional radiative channels and energy–momentum transfer

The accelerating source couples to the complete medium, not only to the desired light branch. The calculation must separately determine

$$P_{u_T}, \quad P_{\text{odd}}, \quad P_{\text{even}}, \quad P_{u_L}, \quad P_{\text{bulk}}, \quad P_{\chi},$$

where P_{u_T} includes both transverse light polarizations; the odd and even terms cover the electric-normal and thickness/order sectors; P_{u_L} is in-plane longitudinal emission; P_{bulk} covers density or sound modes in the surrounding phase; and P_{χ} covers conversion or internal-reservoir response.

Additional branches discovered by the complete spectrum must be added rather than folded into one residual loss number.

For every channel, the project must establish the receiving dispersion, source Fourier support, mode-projector overlap, interface overlap, outgoing boundary condition, energy or norm sign, and emitted flux. Kinematic intersection with a continuum is not enough to prove radiation, but a desired symmetry suppression must be derived rather than asserted. Harmonics and sidebands generated by an intrinsically periodic supported throat may radiate even when the mean source is slow. A dissipative branch may absorb power without a propagating far-zone wave. Both effects belong to the closure test.

The desired ordinary regime requires transverse emission to dominate in the specified electromagnetic process while every extra channel is absent, symmetry suppressed, screened, threshold closed, or quantitatively acceptable. The proposal does not require exact Maxwell theory. It permits calculable longitudinal, bulk, order, or preferred-medium departures, but they must be small enough in the declared regime and remain consequences of the one frozen medium. An extra channel cannot be suppressed by reducing a coupling that the static-electric or magnetic calculation already fixed unless the change is propagated through all three sectors.

This stage fails as electromagnetic closure if the static, magnetic, and radiative normalizations cannot be tied to one source; if acceleration does not couple correctly to both transverse light polarizations; if energy and momentum cannot be balanced; if the electric carrier's dynamic branch is inconsistent; or if unavoidable extra radiation is unacceptable. A narrower failure may reject one source profile, one speed ordering, or one candidate parent response. The handoff must record the tied current and coupling, normalized transverse emission kernel, complete channel powers, frequency and angular dependence, retarded poles, and every speed or threshold that constrains the common medium.

5.6 Passive response, inertia, extra modes, and drag

5.6.1 Active, passive, and inertial roles

The far-field program must not use the word *mass* as though one scalar coefficient had already been derived. Active gravitational strength, passive response, and inertial resistance are different measurements of the complete object. They may ultimately obey universal low-energy ratios, but their equality cannot be imposed by one normalization.

Active strength is a source-side quantity. Once the local signed gravity coefficient, matching prescription, and calibration are fixed, an observer may parameterize the local source by an active mass. That definition records the strength of the derived environmental field; it does not identify the source with trapped-wave energy, constituent rest mass, aperture throughput alone, or any other single microscopic quantity. Return may make the effective active source scale dependent outside the local matching regime.

Passive response belongs to a later test throat placed in the already calibrated external field. In the low-frequency limit, write

$$F_i^{\text{test}}(\omega \rightarrow 0) = \mathcal{M}_{ij}^{\text{passive}}(0)g_{\text{eff},j} + \dots.$$

The source field is held fixed while the response of different throat species is calculated. On the ordinary stable branch, the symmetric quasistatic response must be positive definite. A reference throat may set units, but the calibration may not be altered for each later species to force equal acceleration. Tensor, frequency, field-strength, and environmental corrections must remain visible outside the declared universal regime.

Inertial response is the resistance encountered when the complete dressed throat is accelerated. The relevant object includes the support mode, throat shape, brane deformation, near-field flow, pressure and density response, conversion region, and reservoir dressing. Two broad reversible contributions are anticipated. The trapped standing pattern may need to retune its counterpropagating components, shift phases and nodes, or alter its resonant geometry. The surrounding material configuration must also be rebuilt as the throat velocity changes. Energy and momentum stored in those adjustments can produce an added-inertia-like response without identifying observed mass with a sum of microscopic constituent masses.

For a closed reversible branch, the low-speed inertial tensor may be extracted from the total momentum of the complete solution,

$$M_{ij}^{\text{eff}} = \left. \frac{\partial P_i}{\partial v_j} \right|_{\mathbf{v}=0},$$

with all field and near-zone contributions included. For a relaxational or mixed branch, the response is frequency dependent,

$$F_i(\omega) = \left[-\omega^2 M_{ij}(\omega) - i\omega \Gamma_{ij}(\omega) + \dots \right] \delta x_j(\omega).$$

The real, reactive part describes stored momentum and effective inertia; the absorptive part describes damping, conversion lag, heat, or reservoir excitation. Both must follow from one closed retarded system. An unexcited internal reservoir must be passive: a small periodic perturbation may deposit zero or positive net work into the complete throat-plus-reservoir system, but it may not extract unlimited work from an unprepared bath.

Only after positive ordinary active, passive, and inertial behavior is established are universality ratios meaningful. The target low-energy tests have the schematic form

$$\frac{M_{\text{passive}}}{M_{\text{inertial}}} \longrightarrow \eta_{\text{PI}}, \quad \frac{M_{\text{active}}^{\text{local}}}{M_{\text{inertial}}} \longrightarrow \eta_{\text{AI}}$$

in a weak-field, low-velocity, low-frequency, homogeneous environment and a valid local gravity matching regime. The constants may be normalized only after the source and response conventions are frozen. Universality of a negative, singular, unstable, or active response does not count as success. Nor would these ratios alone establish the full geometric equivalence principle of GR; they are the model's operational low-energy universality requirements.

The far-field stage can determine what source and response kernels are required, which background

modes contribute, and how species-dependent source residues would enter. It cannot complete the mass test until moving supported-throat families exist. Its handoff to the throat program is therefore explicit: each solved species must later supply active source data, passive response, inertial response, and their dependence on internal state without changing the common background calibration.

5.6.2 Complete mode inventory, thresholds, leakage, and friction

A preferred material frame makes the complete spectrum part of the force phenomenology. The model contains, or may contain after full diagonalization, transverse guided light, in-plane longitudinal motion, odd electric/normal response, even thickness response, bulk sound or density branches, order modes, interface modes, and reservoir relaxation. Gravity may involve more than one of these branches. Their characteristic speeds are not automatically equal: c_γ , the bulk sound speed c_s , a possible electric speed c_E , and the longitudinal speed c_L arise from different combinations of common coefficients. Cone alignment may emerge, remain approximate, or fail. Each possibility changes leakage, radiation thresholds, and the observable preferred-medium scope.

On a reversible homogeneous branch, uniform motion can emit into any coupled mode satisfying the kinematic resonance condition. In an isotropic background, a useful first diagnostic is

$$v_c = \min_a \inf_{k>0} \frac{\omega_a(k)}{k},$$

where a runs over every physical branch with nonzero throat coupling. In an anisotropic or defect background the threshold is directional and must use $\omega_a(\mathbf{k}) = \mathbf{k} \cdot \mathbf{v}$. A quadratic dispersion $\omega \propto k^2$ has vanishing ω/k at long wavelength, so a nonzero drag-free interval would then require symmetry suppression or a vanishing source residue rather than merely a small velocity. A branch may also be kinematically open yet dark because the relevant projection vanishes; the source overlap and outgoing flux must be calculated.

Dissipative or mixed branches require an additional test. Absence of a propagating wave does not imply absence of drag. The low-frequency friction, order-relaxation, diffusion, and conversion kernels must vanish or remain within the declared regime for uniform translation. A long-range static electric response with a diffusive pole, for example, may leave a wake even when no on-shell wave is emitted. The complete response should separate reversible added inertia, acceleration radiation, and steady friction instead of fitting them with one effective mass.

Motion may also change the source itself. For an isotropic throat, scalar throughput and conversion corrections begin naturally at even powers of speed, for example

$$Q_A(v) = Q_{A0} \left[1 + \alpha_A \frac{v^2}{c_*^2} + O(v^4) \right], \quad A \in \{n, \chi\}.$$

The coefficients are outputs of the moving-throat family. Any resulting change in the signed gravity source $\overline{Q}_g(r; v)$ is a velocity-dependent gravity effect, not inertial mass or electromagnetism. Likewise, velocity-dependent electric coupling, polarization, slab thickness, or reservoir loading would be a preferred-medium prediction that must be propagated through the magnetic and radiation sectors.

The full audit therefore combines static range with dynamic risk. It asks which modes are long ranged, which are screened, which are constrained, which can carry energy away, which create low-frequency lag, and which couple to a stationary, moving, or accelerating throat. It must include leakage from the brane into bulk continua, conversion among transverse and longitudinal modes, odd and even radiation, and reservoir dissipation. No unwanted branch may be omitted because it is not part of the desired analogy.

The selected common-medium branch fails this stage if an ordinary stable throat would possess a non-positive passive or inertial direction; if the retarded response is active for an unexcited reservoir; if low-energy mass ratios are unacceptably species dependent; if any unavoidable coupled mode gives a vanishing or too-low critical velocity; if uniform motion has unacceptable friction; or if the required characteristic-speed relations destroy light confinement, Coulomb dynamics, or gravity response. A less severe result may instead narrow the model to a preferred-medium regime with explicit bounded departures. The scope of the conclusion depends on the calculated magnitude and whether the conflict persists throughout the allowed common parameter region.

5.7 Far-field requirement summary

The force program does not end by assigning a separate pass label to gravity, electricity, magnetism, radiation, and inertia. It ends by exporting one linked set of source, response, and spectral requirements. The principal handoff is summarized in table 5.1.

Several shared quantities appear repeatedly and must be represented only once in Candidate Parent System V1: the slab thickness and background profiles; the order branch and reservoir variables; the complete constrained static and retarded operators; the transverse light normalization; odd and even source profiles; the oriented current; the independent couplings; and the boundary and environmental ensembles. The force requirements therefore constrain one another before a throat is solved. A parameter that screens an unwanted even force may change the photon helper. An electric kinetic term may open extra radiation. A speed choice that reduces drag may move light or longitudinal modes into a bulk continuum. A distributed porous-brane S_{leak} return law that produces the desired local gravity source may alter the reservoir state used by every other sector.

The common-medium synthesis in chapter 6 will place these outputs beside the light requirements and test their intersection. A robust region must satisfy every hard structural gate and keep every bounded departure inside its declared range at the same parameter point or throughout a single declared region. A fragile point lying on a pole collision, continuum threshold, sign change, or vanishing source residue is not equivalent to a robust capability region, even if one numerical evaluation appears favorable. An empty intersection rejects the candidate combination rather than authorizing a private repair for the sector that failed.

Even a successful far-field intersection remains conditional until one frozen parent system produces the stable slab and complete spectrum and then rederives all of the responses above. It also remains conditional on the later throat program: actual supported objects must realize the provisional source profiles, conserved current, universal couplings, positive response, and acceptable motion. The purpose of this chapter is to make that later solve decisive. The throat may select among permitted source families, but it may not change what gravity, electricity, magnetism, radiation,

Table 5.1: Principal far-field handoff to the common-medium synthesis.

Category	Required output	Principal rejection mode
Gravity source	Probe-independent source reduction; reflection-even signed coefficient; attractive inverse-square local regime; distributed porous-brane S_{leak} return and matching map; one frozen calibration	Wrong sign or parity, no usable local regime, species-dependent calibration, or unacceptable screening, overshoot, or retarded response.
Electric static sector	Complete odd/even Hessian blocks; healthy source-coupled odd $\kappa_E k^2$ branch; correct physical force ensemble and sign; screened or bounded additional forces	Wrong range or sign, ghost or instability, or an unavoidable extra long-range branch.
Charge universality	Nonnegative one-throat magnitudes g_a , bounded factorization corrections, conserved oriented count/current, and projected Gauss-like relation	Independent normalization by species, no conserved current, or large nonfactorizing corrections.
Electric dynamics	Propagating, relaxational, mixed, auxiliary, or constrained classification of the full odd retarded block, with poles, residues, damping, passivity, and zero-frequency closure	Static Coulomb behavior necessarily entails unacceptable radiation, drag, dissipation, strong coupling, or constraint inconsistency.
Magnetism	$O(v)$ orientation-odd transverse source, required tensor and falloff, and normalization inherited from static charge and the light operator	Independent magnetic coupling, wrong reversal law, or domination by an orientation-even wake.
Radiation	$O(a)$ transfer into both transverse light polarizations, shared current and coupling, radiation reaction, and channel-resolved energy-momentum balance	Missing tied transverse emission, failed conservation, or excessive odd, even, longitudinal, bulk, or conversion radiation.
Mass response	Positive active source, positive passive and inertial tensors, retarded passivity, and bounded low-energy species variation of the required ratios	Wrong-sign or active response, singular inertia, or failed low-energy universality.
Speeds and drag	Complete characteristic-speed, continuum, source-overlap, critical-velocity, leakage, and low-frequency friction map	Unavoidable low-threshold emission, vacuum drag, cone incompatibility, or uncontrolled preferred-medium effects.

inertia, and drag were required to mean.

Cross-Sector Integration and the Common Medium

The light and far-field chapters do not define several independent effective models. They define a linked set of conditions that one parent medium must meet. This chapter performs the transition from those conditions to an actual realization attempt. It normalizes the sector outputs in one Medium Capability Matrix, identifies the quantities that several sectors share, tests whether a common admissible region exists, and states what must be frozen in Candidate Parent System V1. The candidate is then required to produce one stable brane and one complete response spectrum from which the light and far-field results are rederived.

The compatibility burdens collected here are not established contradictions. The companion cross-sector review found no present no-go theorem. Each burden becomes a falsification gate only when a frozen system shows that a mandatory capability is impossible or necessarily entails a forbidden consequence. The central risk is therefore not an already-proved inconsistency, but the possibility that requirements which are separately plausible have an empty intersection when imposed on one medium.

6.1 The Medium Capability Matrix

The Medium Capability Matrix is the common interface between requirement discovery and model construction. It is not a narrative summary and it is not a scorecard. Each row identifies one controlled physical or mathematical object, the property required of it, the regime in which that property is required, the sectors that use it, and the evidence needed to decide whether the requirement is satisfied. Two sectors that refer to the same mode, coefficient, source, or normalization must point to the same row rather than introducing parallel versions of it.

The matrix must preserve four different kinds of entry. First are *candidate choices*: the field inventory, dynamical branch, operator basis, symmetries, independent coefficients, and boundary or reservoir ensemble. Second are *background outputs*: the slab profile and thickness, guided normal profiles, mode compositions, eigenvalues or retarded poles, continuum thresholds, and physical normalizations. Third are *provisional source contracts*: the parity, multipole content, continuity law, and symbolic form factor required of an eventual throat. Fourth are *later nonlinear outputs*: actual throat source magnitudes, universality, inertial and passive response, pair topology, and the

converged reservoir state. Keeping these classes separate prevents an unknown throat or global quantity from being promoted to an adjustable parent parameter.

A compact matrix schema is shown in table 6.1. The filled matrix remains a working companion to the proposal; the proposal needs only enough structure to make the common-medium decision unambiguous.

Table 6.1: Compact schema for the Medium Capability Matrix.

Entry class	Controlled content	Cross-sector use
Background and branch	Arena, fields, conservative, relaxational, or mixed dynamics, finite-slab profile, shear support, constraints, and reservoir variables	Establishes the common host on which every mode, source, force, and lifetime calculation is performed.
Mode or response branch	Field composition, reflection parity, normal profile, static stiffness, dispersion or retarded pole, gap, continuum placement, norm or energy sign, damping, leakage, and observable residue	Connects guided light, the electric odd and even sectors, longitudinal and bulk response, radiation, throat support, critical velocity, and drag.
Source and coupling	Odd and even source vectors, form factors, current or constraint identity, one-throat coupling magnitude, gravity-source reduction, and mode overlap	Connects static forces to moving-source response, acceleration radiation, universality, and the later replacement of provisional sources by solved throats.
Observable and ensemble	Force or flux extraction, equilibrium, periodic, or driven ensemble, outgoing conditions, one-time calibration, and declared physical regime	Prevents a static energy, retarded response, mode normalization, or calibration from being changed between sectors.
Decision record	Exact gate, bounded departure, or scope choice; metric and domain where needed; uncertainty; current evidence; failure scope; and dependencies	Converts qualitative words such as “small,” “long lived,” and “universal” into a conjunctive common-medium decision.

For every background eigenmode, the matrix should record enough information to serve both the low-wave-number force audit and the finite-wave-number light program. A label such as “longitudinal mode” is insufficient. The same row must state its actual mixture of displacement, interface, density, order, bulk, and reservoir variables; its parity; its profile across the slab; its static and dynamic behavior; its source overlaps; and whether it is desired in one sector but unwanted in another. The light program may use a reflection-even mode as a packet helper while the electric program requires the same branch to avoid an additional long-range force. That is one shared constraint, not two independent design choices.

The matrix also records the difference between an exact structural gate and a bounded physical departure. Exact constituent conservation, a nonredundant branch closure, bounded physical energy

or branch-appropriate stability, the absence of ghosts, the correct number of freely selectable light polarizations, a legitimate force construction, and one shared normalization cannot be traded against performance elsewhere. Leakage, resonance lifetime, weak dispersion, extra radiation, species variation, return screening, and low-frequency drag may instead be bounded over a declared regime. Detailed numerical budgets and evidence paths remain in the capability matrix and derivation ledger rather than being reproduced here.

6.2 Shared fields, coefficients, speeds, sources, and constraints

The common-medium test begins by identifying quantities that must have one meaning everywhere. The first group belongs to the host material: the background profiles $n_0(w)$ and $\chi_0(w)$, the selected thickness H_{br} , the ordered-state shear law, the equation of state, and the chosen conservative, relaxational, or explicitly mixed order dynamics. A sector may not use a conservative free-energy Hessian while another silently treats the same order variable as dissipative without the associated reservoir. Different branch choices define different candidates because they change the phase space, constraints, conserved quantities, force formalism, damping, entropy ledger, and particle-longevity problem.

The second group is the complete constrained response. Where an equilibrium reduction exists, the static calculation uses one physical Hessian $\mathcal{H}(\mathbf{k})$. Dynamics uses one retarded operator $\mathcal{O}_R(\omega, \mathbf{k})$, including every internal variable needed by the selected branch. Their parity blocks, constraints, poles, branch cuts, continua, norm signs, damping rates, profiles, and residues are shared data. The transverse speed c_γ , longitudinal speed c_L , bulk compressional speed c_s , and any electric-sector speed or relaxation scale are therefore outputs of one spectrum. They cannot be assigned separately to obtain light confinement, Coulomb range, gravity propagation, and low drag.

The third group is the source hierarchy. The reflection-odd and reflection-even source vectors, the oriented density and current, the leading nonnegative electric coupling magnitude g_a , the gravity source-side reduction, and the normalization of outgoing transverse modes must retain one parent definition. A provisional far-field source may leave its amplitude and form factor symbolic, but its transformation law and continuity assumptions must be explicit. The later throat program must derive those quantities. It may select among source families permitted by Candidate V1; it may not alter the background operator or introduce a new electric, magnetic, radiative, or gravity coupling after seeing the result.

Boundary and ensemble data are shared as well. Reflection covariance requires a specified transformation of the environmental data. An equilibrium pair, a conservative periodic pair, and a driven dissipative pair do not automatically share one interaction potential. Outgoing-wave conditions, fixed source or fixed boundary-value ensembles, support-mode action or phase conditions, and reservoir asymptotics must therefore be frozen with the candidate. Otherwise a force sign or lifetime could be changed by changing the problem rather than the medium.

These shared objects create direct cross-sector tensions. Increasing a thickness-mode gap may screen an unwanted even electric force while weakening a photon helper. A higher-gradient term may aid packet localization while adding vacuum dispersion or a low-threshold branch. Giving the odd Coulomb carrier ordinary inertia may produce a clean wave but also extra radiation; making it

relaxational may replace radiation with lag, friction, and reservoir cost. Locking characteristic speeds may improve an emergent common cone while placing a guided mode at a bulk continuum threshold. A distributed porous-brane S_{leak} return law that supplies the desired local gravity regime changes the asymptotic reservoir seen by the slab, light, and throat. The purpose of the shared-quantity map is to make these consequences unavoidable.

6.3 Major cross-sector compatibility gates

Seven architecture gates collect the largest known ways in which separately reasonable requirements may conflict. They summarize the detailed companion review rather than replacing it. Each gate begins as a present burden. An adverse calculation normally rejects a branch or Candidate V1 first. It becomes an architecture-level result only if the obstruction persists throughout the admitted common-medium class or can be avoided only by prohibited retuning or an unsupported new mechanism.

Table 6.2: Proposal-level summary of the seven cross-sector architecture gates.

Gate	Common burden	Earliest decisive test and initial adverse scope
1. Frozen candidate	The broad ontology still permits more than one dynamical branch, operator basis, shear realization, higher-gradient completion, and reservoir closure. Results cannot be combined while sectors choose different members of that menu.	Freeze Candidate V1, then solve its finite slab and complete constrained response. Failure initially rejects V1; it threatens the architecture only if no allowed frozen candidate can realize the contracts.
2. Odd Coulomb dynamics	A healthy reflection-odd branch with leading k^2 static stiffness can produce a Coulomb tail, but that fact does not determine whether the branch propagates, relaxes, or is a genuine constrained variable.	Derive the complete odd static and retarded blocks, constraints, poles, residues, radiation, passivity, and low-frequency friction. Static success does not pass if every completion creates a ghost, strong coupling, unacceptable radiation, drag, or independent electromagnetic normalizations.
3. Speeds and confinement	Relations among c_γ , c_L , c_s , electric response, and slower branches affect guided-mode localization, continuum thresholds, gravity response, mode conversion, and the critical velocity of a moving throat.	Compute the full spectrum, essential continua, source and interface overlaps, leakage widths, and directional all-branch critical velocity. Exact cone locking may survive through a derived decoupling; distinct speeds may survive only as an explicitly bounded preferred-medium analog.
4. Universality and discreteness	Distinct throat species must coexist with sufficiently universal unit electric coupling, positive and universal low-energy mass ratios, and a conserved oriented count or current. Continuous internal moduli may instead produce arbitrary charge, size, throughput, or mass.	Continue solved throat families and measure source residues, response ratios, topology, and sensitivity to internal parameters. Failure initially rejects the proposed particle family or its charge and mass interpretation, not the already-derived background response.

Gate	Common burden	Earliest decisive test and initial adverse scope
5. Photon binding and linearity	The nonlinearity and dispersion strong enough to bind one finite packet also affect ordinary light speed, vacuum dispersion, polarization, wakes, collisions, even-sector forces, and throat support.	From the same projected V1 coefficients, test boundedness, scaling, collapse, speed shift, helper radiation, and preliminary packet interactions. Failure of the shared binding window rejects the self-bound photon ontology before it rejects a viable linear guided-light sector.
6. Localized drainage, distributed return, and reservoir closure	Persistent one-way throat throughput and ordered-to-bulk conversion link throat longevity to energy, momentum, entropy, drag, gravity screening, and depletion of the common bulk reservoir. Compensating material return is a distinct process: bulk reorganizes into brane state across the porous brane through the positive bulk-to-brane contribution to S_{leak} .	Iterate the local supported-throat problem with the global bulk-reservoir and distributed porous-brane return problem to a stable branch-consistent fixed point. Failure rejects the selected conservative, dissipative, or mixed throat branch; it threatens the architecture only if no branch can close.
7. Contact and topology	Cancellation of the leading odd electric disturbance, projected coincidence, four-dimensional core contact, topology change, and annihilation are distinct. Same-species partners must also be distinguished from opposite charges of different internal or composite species.	Solve both orientations, continue two-core families through graph breakdown, and evolve conserving nonlinear collisions. Failure first rejects a simple annihilation mechanism or species map; the particle-antiparticle sector fails only if no conserving channel exists.

The first three gates can be attacked substantially at the background and linear-response stage. The photon gate begins there and continues through the finite-amplitude light calculation. The universality, reservoir, and topology gates cannot be finally closed without solved throats and global evolution, but they constrain Candidate V1 before that work begins. The candidate must contain an admissible conservation structure, nonlinear field content, branch-specific reservoir, and topology capable of posing those tests. A later failure may reveal that V1 lacks the required enforcing structure; it may not be repaired by adding one only to the failed sector without declaring a new candidate version.

6.4 The common viable-region test

Let λ denote the independent continuous parameters of one declared candidate architecture, with discrete branch, symmetry, operator, and boundary choices carried as explicit labels rather than hidden coordinates. For each mandatory requirement i , let Λ_i be the subset on which its exact gate closes or its bounded departure remains inside the declared admissible set. The common contract-feasible region is

$$\Lambda_{\text{contract}} = \bigcap_{i \in \mathcal{I}_{\text{mandatory}}} \Lambda_i.$$

Equivalently, at the architecture level the principal burdens may be displayed schematically as

$$\Lambda_{\text{architecture}} = \Lambda_{\text{slab}} \cap \Lambda_{\text{guided light}} \cap \Lambda_{\text{Coulomb}} \cap \Lambda_{\text{electric dynamics}} \\ \cap \Lambda_{\text{throat universality}} \cap \Lambda_{\text{photon packet}} \cap \Lambda_{\text{reservoir}} \cap \Lambda_{\text{topology}}.$$

Before throats are solved, some of these sets express necessary structural conditions rather than completed existence results. A nonempty $\Lambda_{\text{contract}}$ therefore authorizes selection of a realization attempt; it does not yet prove that the background, throats, or reservoir fixed point exist. Candidate V1 must convert the provisional intersection into a realized one by producing the objects used to evaluate it.

Mandatory requirements are conjunctive. No weighted score permits a long photon lifetime to compensate for failed charge conservation, a clean Coulomb tail to compensate for a ghost, or an attractive gravity regime to compensate for an external energy source disguised as an internal reservoir. Optional or stronger capabilities may be reported separately, and a failure may reduce the model's scope, but they cannot pay for a failed mandatory gate. Every point reported as common must use the same parent version, branch, background, source conventions, and shared parameter values.

The geometry of the surviving set is itself a scientific result. An empty set means that the tested combination cannot be realized. An isolated point or singular locus may indicate a special exact solution but provides no robustness without a derived protection. A narrow band close to a continuum threshold, pole collision, sign reversal, instability, collapse boundary, or vanishing source residue is fragile. Several disconnected components represent distinct candidate branches and must not be blended. A robust region contains a finite neighborhood in which all hard gates remain closed and every bounded departure retains a margin larger than the relevant numerical and model uncertainty. Perturbing shared coefficients, boundary data, regulator choices, and retained higher-order terms within their declared uncertainty should not immediately reverse the decision.

The matrix should therefore report not only passing points but the limiting requirements, distances to decision boundaries, uncertainty on those margins, and which components of the intersection remain conditional on later throat or reservoir work. A broad exploratory scan may locate promising regions or no-go surfaces. The final claim, however, is the common region after one candidate and its evaluation rules have been frozen.

6.5 Freezing Candidate Parent System V1

If the contract-feasible intersection is nonempty, the project selects one complete realization for a full test. Candidate Parent System V1 is not the entire architecture and is not a menu from which each sector may choose. It can be represented schematically as

$$\mathfrak{C}_1 = (\mathcal{A}_1, \Phi_1, \mathcal{E}_1, \mathcal{R}_1, \Sigma_1, \lambda_1 \in \Lambda_1, \mathcal{Q}_1, \mathcal{O}_1),$$

where the entries denote the parent arena; complete field content; equations, constitutive laws, and operator basis; internal reservoir closure; symmetries and constraints; independent parameter point or domain; boundary, source, and ensemble rules; and operational maps used to extract observables. The explicit freeze must state the maximum derivative order, any allowed nonlocality or memory, time-reversal and reflection properties, the physical constrained space, asymptotic and outgoing conditions, and the one-time calibration choices.

The field inventory includes every fundamental, retained effective, auxiliary, constrained, and reservoir variable needed by the selected branch. The equation set must specify whether the order sector is conservative, relaxational, or mixed and must close the corresponding material, momentum, energy, and, where applicable, entropy transfers. The operator-basis freeze is as important as the numerical parameter freeze. A term omitted from V1 is absent for V1; it cannot reappear later as a photon-only, electricity-only, or throat-only repair.

Candidate V1 may retain a declared common parameter domain when a family rather than one number is being tested. It may not retain materially different field contents or branch choices under one name. Alternative completions remain separate candidates. Simplicity is a discriminator only after mandatory capabilities are represented: omitting a required degree of freedom is not a virtue, while adding speculative operators that no contract requires weakens the meaning of later failure.

The freeze also identifies what remains to be derived. The stable thickness, mode profiles, electric eigenvector, throat radius, source magnitudes, universality, packet solution, distributed porous-brane S_{leak} return profile, and converged reservoir state are not independent knobs merely because they are not yet known. They are outputs of V1. Once obtained, they become inherited inputs for downstream calculations with their regimes and uncertainty attached.

Freezing does not certify that V1 is correct. It makes the test definite. If a later calculation exposes a missing assumption or an inconsistent operator, the project may define V2. That revision must state the change and reopen every real dependency. Results derived from V1 remain historical evidence about V1; they cannot be combined with V2 outputs as though the parent system had never changed.

6.6 Stable brane, complete spectrum, and far-field rederivation

Candidate V1 first faces the least expensive decisive realization test: can its own equations produce the host medium assumed by the capability contracts? The background solve must yield a finite ordered slab with selected thickness, appropriate reflection properties, positive shear support, bounded or branch-appropriate stable energy, and stable or sufficiently metastable interfaces. The background profiles and asymptotic reservoir data must solve the same V1 equations and boundary conditions that will be used downstream.

The next task is the complete constrained linear response about that slab. The calculation must inventory every physical transverse, longitudinal, interface, density, phase, order, bulk, electric, conversion, and reservoir branch. For an equilibrium sector it derives the physical static Hessian

and its parity blocks. For dynamics it derives the full retarded operator. The output includes constraints and auxiliary variables; eigenvectors or left and right response modes where appropriate; static stiffnesses; poles, cuts, and continua; norm and energy signs; characteristic speeds; gaps and screening lengths; damping and leakage; normal profiles; source and interface overlaps; flux normalizations; and observable residues. A sector-specific truncation is acceptable only after this full operator shows that the omitted variables decouple in the stated regime.

This one background calculation supplies the common inputs to several tests at once. It must recover a healthy two-polarization guided transverse branch or record the failure of the selected light branch. It must identify the longitudinal and bulk thresholds that control helper radiation and drag. It must show whether a reflection-odd source-coupled eigenbranch has the required k^2 static stiffness and whether its retarded completion is propagating, relaxational, mixed, auxiliary, or constrained. It must classify the even thickness and order response, every additional long-range force, and the modes available for gravity response, throat support, radiation, and uniform-motion emission.

The far-field contracts are then rerun on this actual background. Guided-light profiles, speed, dispersion, leakage, and nonlinear projection coefficients are computed from V1 rather than imported from a provisional action. Gravity uses the V1 response and source-side variables with one frozen calibration and a distributed porous-brane S_{leak} return ensemble. Static electricity uses the complete V1 odd and even blocks, the physical boundary ensemble, and one source normalization. Magnetism and acceleration radiation use the same oriented current, coupling, transverse mode norms, and retarded response. The mass and drag audits use the same spectrum, reservoir, and source conventions. Every unwanted odd, even, longitudinal, bulk, and conversion channel remains in the accounting.

At this stage the localized source may still be a controlled placeholder, but it is now a source class permitted by V1 and coupled to V1's actual normalized modes. The later throat solve must replace its amplitude, profile, conservation law, and universality assumptions. Thus a successful rederivation establishes a realized *common far-field medium*, not yet a realized particle sector. A candidate that has a stable slab but requires different parameter points for guided light, Coulomb response, acceptable dynamics, and low drag fails the common-medium test just as surely as a candidate with no stable slab.

A useful realized region may be written

$$\Lambda_{\text{far}}^{(1)} = \Lambda_1 \cap \Lambda_{\text{slab}}^{(1)} \cap \Lambda_{\text{spectrum}}^{(1)} \cap \bigcap_{S \in \mathcal{S}_{\text{far}}} \Lambda_S^{(1)}.$$

Only if this set is nonempty, with every mandatory result evaluated on the same V1 background and parameter point or region, does the program advance to the supported-throat and particle work in chapter 7.

6.7 Decision outcomes of the common-medium phase

The common-medium phase can end with several scientifically distinct outcomes. An empty contract intersection means that the tested capability combination does not justify a Candidate V1. It becomes an architecture failure only after the admitted common-medium class has been examined broadly enough to show that the obstruction is structural rather than peculiar to one realization.

A nonempty but isolated or fragile intersection justifies a focused realization attempt, not a robust success claim. Candidate V1 may be frozen to determine whether the apparent point survives the full background, spectrum, convergence, and perturbation tests. Its proximity to every threshold must remain part of the result. Disconnected viable components should be treated as separate candidates rather than combined into a single flexible system.

A contract-feasible Candidate V1 can still fail because it has no selected stable slab, an unhealthy constrained spectrum, a wrong force sign, an unacceptable extra channel, incompatible characteristic speeds, or an empty realized far-field region. That is a controlled candidate failure and is one of the principal intended outputs of the program. It identifies which capability combination could not be realized by the frozen equations.

A weaker sector may also survive after a stronger claim fails. A healthy guided linear-light branch may remain even if every shared self-bound photon packet fails. A locally inverse-square gravity analog may remain while a larger-scale return behavior limits its scope. Such outcomes must be declared as partial analogs, not averaged into either complete success or complete failure. Whether a stronger capability is mandatory for V1 must be stated in the candidate's scope before its decisive test.

The positive exit condition for this chapter is deliberately limited: one versioned parent system produces a stable common background and complete healthy response and rederives all mandatory light and far-field capabilities in one shared region. That result authorizes the nonlinear throat program. It does not establish particle existence, charge discreteness, universal mass response, annihilation, or the global reservoir fixed point. Those deferred gates remain capable of rejecting the particle branch or Candidate V1 later. The value of the common-medium phase is that either outcome—a surviving common background or a controlled incompatibility—has a precise meaning without sector-specific retuning.

Beyond the Far Field: Throat and Particle Program

The common-medium phase in chapter 6 ends, at best, with a stable finite brane, a complete background response, and a realized far-field region for one frozen Candidate Parent System. That is not yet a particle theory. The localized sources used in the far-field calculations still have to be replaced by solutions of the parent equations whose geometry, internal support, transport, conversion, couplings, response, and reservoir loading all arise from the same candidate medium.

This chapter outlines that nonlinear program. It begins with one supported throat, continues through species and orientation partners, constructs moving and accelerating families, and then studies two-throat interactions from the far field through finite-thickness contact. The final stage closes the feedback among each local one-way throat drain, the global bulk reservoir, and the separate distributed S_{leak} return across the porous brane. No new particle-only mechanism is introduced at this stage. The purpose is to determine whether the structures permitted by Candidate V1 actually realize the source classes, signs, normalizations, stability, and conservation assumptions used in the earlier chapters.

7.1 One-throat existence, support, and stability

A candidate particle is a finite nonlinear throat through the ordered slab, not a point source placed on an otherwise fixed background. Its solution must include the full parent fields in $(\mathbf{x}, w)$, the deformed interfaces where those interfaces remain meaningful, the surrounding density and flow, the local order conversion region, the trapped support field, and the asymptotic reservoir data. The apparent mouth seen by a brane observer is only one projection of this complete object.

The first calculation is a spectral seed on a provisional throat geometry. The transverse support field must solve the same variable-coefficient shear operator that governs the light sector, now on a nonuniform defect background. Schematically,

$$\mathcal{L}_T[\chi_B, n, \text{geometry}] \phi_T = \omega^2 \mathcal{W}_T[\chi_B, n] \phi_T.$$

A true bound seed requires finite weighted norm, finite energy, and acceptable asymptotic decay.

A leaky seed is a different problem: it requires outgoing conditions, a complex pole or equivalent leakage rate, and a derived lifetime. The support region may emerge as a rim, collar, sheath, cavity wall, or another localized pattern of stress. It must be diagnosed from the solved energy and stress distribution rather than imposed as a hard domain in advance. A background light mode that is healthy on the homogeneous slab is not thereby a bound throat mode; the defect spectrum, bulk continua, longitudinal coupling, and order response must be recalculated.

The spectral seed is not the particle. It must be continued to a self-consistent free-boundary or time-periodic solution. In that solution the support mode's cycle-averaged stress holds open the aperture whose shape sets the mode spectrum. The aperture supplies a localized one-way drain, and its ordered-to-bulk conversion loads the global reservoir. Bulk material does not reverse course through the throat. It returns separately by reorganizing into brane state across the porous brane, through the positive bulk-to-brane contribution to S_{leak} . Mechanical backpressure can still couple the reservoir to the local throat boundary data without constituting material return through the throat. The intended closure is therefore

$$\begin{aligned} E_{\text{support}}, \omega_{\text{standing}} &\longleftrightarrow R_{\text{throat, shape}} \longrightarrow \{Q_n^{\text{net}}, Q_n, Q_\chi\}_{\text{localized drain/conversion}} \longrightarrow \{p_{\text{bulk}}, \mathcal{B}_\infty\}, \\ \{p_{\text{bulk}}, \mathcal{B}_\infty\} &\longrightarrow S_{\text{leak}}^{\text{bulk} \rightarrow \text{brane}}(\mathbf{x}, t) \longrightarrow \text{distributed reordering across the porous brane} \\ &\longrightarrow \text{brane background and local throat boundary data.} \end{aligned}$$

Here the signed net constituent flux, the selected positive throughput on an outward-drain branch, and the gross local ordered-to-bulk conversion diagnostic remain distinct. Their relationship is an output of the chosen conservative, relaxational, or explicitly mixed order dynamics. A critical-flow estimate may supply a necessary throughput bound for a fixed mouth and upstream state, but it cannot by itself select the throat radius, establish stability, or turn support energy into observed mass.

The nonlinear continuation must determine the equilibrium or cycle-averaged radius and shape, whether the core is anchored mainly to one interface or spans the slab, the support frequency and localization, the transport and conversion profiles, and every source component exported to the far field. In particular, it must replace the provisional electric and gravity sources with the throat's complete odd and even source vectors, its leading coupling to the Coulomb carrier, and its local source-side transport, pressure, stress, and ordered-to-bulk conversion data. The associated distributed porous-brane S_{leak} return profile is a separate output of the coupled global problem. The gravity field must remain probe independent, and the same one-time calibration used in chapter 5 must be retained.

A converged stationary geometry may still contain a dynamically oscillating support mode. Stability must therefore be tested about the complete solution, not inferred from the seed eigenvalue. A conservative periodic throat requires a Floquet or equivalent Hamiltonian response analysis. A relaxational or mixed throat requires the full retarded linearization, including every internal reservoir variable used to close damping and entropy production. The audit must include translational zero modes, radius and shape oscillations, support-mode perturbations, orientation-changing channels, negative-energy or growing modes, and leakage into transverse, longitudinal, odd, even, bulk, and conversion branches.

The one-throat stage passes only if at least one branch has a finite supported geometry, nonzero branch-consistent throughput or conversion where required, acceptable localization or lifetime, no unphysical instability, and source profiles compatible with the already frozen far-field contracts. Failure of a particular seed, support family, or order-dynamics branch rejects that route. If no allowed branch of Candidate V1 can produce a sufficiently stable supported throat, V1 has a common far-field medium but no particle realization.

7.2 Species structure, orientation partners, and charge universality

A solved throat family is labeled

$$\mathcal{T}_{\alpha,s}, \quad s \in \{+1, -1\},$$

where s is the normal orientation that supplies the proposed electric sign and α contains the orientation-independent species data. Those data may include the support-mode family, core topology and boundary conditions, equilibrium size and shape, transport and conversion structure, circulation or chirality, and any additional conserved or metastable internal configuration. The program must discover which such branches exist; it may not identify species with orientation by definition.

Both orientations of each species must be solved from the same parent equations under reflected environmental data. The complete antiparticle candidate is

$$\boxed{\mathcal{C}_{\text{th}} = \mathcal{I}_{\text{internal}} \circ \mathcal{R}_w},$$

not geometric reflection alone unless the solved branch shows that the internal operation is trivial. Every orientation-odd phase, circulation, chirality, support convention, topological datum, or conversion current must transform consistently, while the required orientation-even scalars remain equal. For an equilibrium branch this includes equal energy under reflected data. For a conservative periodic branch it also includes a consistent map of support-mode action, frequency, and phase and comparison in one declared cycle-averaged ensemble. For a driven or mixed branch, equality of throughput, dissipation, entropy production, lifetime, stationary stress and momentum fluxes, and retarded response may replace or supplement an equilibrium energy test.

Consequently, reversing an electron-like branch produces the leading positron-like candidate, not a proton-like object. A proton-like object would have to occupy another internal branch, carry additional topology, or be a composite throat structure. This distinction is essential both for annihilation and for neutral binding: opposite electric signs do not imply identical orientation-independent cores.

The same solved families must decide whether charge is universal and discrete. If the far-field odd channel factorizes, its one-throat charge may be written

$$q_{\alpha,s} = s g_{\alpha}, \quad g_{\alpha} \geq 0.$$

The orientation supplies all sign, while g_α is a derived residue or coupling magnitude. Ordinary unit-charge species must produce the same leading magnitude to the required accuracy even when their internal structures differ. A common slab thickness is not enough: the core boundary data must be quantized, topologically controlled, selected by a universal branch condition, or sufficiently decoupled from the long-range carrier. Continuous moduli that allow arbitrary charge, radius, throughput, or source strength would expose a major particle-sector problem unless the resulting variation is part of an explicitly bounded analog regime.

A binary label also does not establish additive conservation. The throat family must support a conserved oriented count or current, with pair creation and annihilation changing the local population while preserving the net permitted quantity. Only after that construction and unit-coupling universality are derived can a conventional long-range charge such as $Q_{\text{eff}} = g_0 N_s$ be assigned. The species study therefore requires more than one attractive solution: it must continue several internal branches, measure their source residues and response coefficients, identify any discrete invariants, and determine whether the antiparticle map and charge law survive across the family without retuning.

7.3 Moving throats, inertia, passive response, and radiation

A stationary throat is next continued into a family with small uniform velocity, external forcing, and controlled acceleration. The moving object is not a rigid mouth translated through a passive medium. Its standing mode must retime and rephase; its throat shape, near-field flow, pressure, density, order profile, and local conversion region must be rebuilt, together with the surrounding distributed porous-brane S_{leak} return disturbance. The response of that complete dressed configuration is the proposed origin of inertia and the source of several observable effects that must remain separately identifiable.

For a closed reversible branch, the low-velocity inertial tensor can be extracted from the total momentum of the complete solution,

$$M_{ij}^{\text{inertial}} = \left. \frac{\partial P_i}{\partial v_j} \right|_{\mathbf{v}=0},$$

where P_i includes the support field, geometry, near medium, order sector, and any reversible reservoir momentum. A relaxational or mixed branch instead requires a frequency-dependent response of the form

$$F_i(\omega) = \left[-\omega^2 M_{ij}(\omega) - i\omega \Gamma_{ij}(\omega) + \cdots \right] \delta x_j(\omega),$$

with the damping, heat, entropy, and internal reservoir variables included in the same closed system. The ordinary branch must have a positive symmetric quasistatic inertial response. Its retarded response must also be passive for an unexcited reservoir: a small periodic perturbation may store or dissipate work, but it may not draw unlimited net power from an unspecified background.

Passive gravitational response is a different calculation. The source-side gravity field and its calibration are fixed before the test throat is chosen. The moving-solution family must then

determine the low-frequency coefficient that maps that external field into force or acceleration. Positive response is a separate gate, followed by the question of whether the low-energy ratios among local active source strength, passive response, and inertial response are sufficiently universal across species. A calibration that absorbs species-dependent response would not establish universality.

Motion also separates three effects that can otherwise be confused. The sign-dependent transverse response proportional to the oriented current is the electromagnetic magnetic sector. Sign-independent rebuilding of the surrounding medium contributes to inertia or wake formation. A speed-dependent change in throughput, conversion, or the signed gravity source is a preferred-medium gravity effect. All three may occur, but they must be extracted from different parity and response components of the same solution.

Acceleration then tests dynamic electromagnetic closure. The same conserved oriented current and the same leading coupling that produced the static electric residue must normalize the $O(v)$ magnetic response and the $O(a)$ transfer of energy and momentum into both transverse light polarizations. The full far-zone calculation must simultaneously measure

$$P_{uT}, \quad P_{\text{odd}}, \quad P_{\text{even}}, \quad P_{uL}, \quad P_{\text{bulk}}, \quad P_{\chi},$$

rather than reporting only the desired transverse radiation. Radiation into the odd electric branch, even thickness response, longitudinal mode, bulk sound, or order and reservoir channels is a prediction of Candidate V1 and may constrain particle lifetime, acceleration response, and electromagnetic accuracy.

Uniform motion supplies an equally important no-drag test. In a reversible branch, every mode coupled to the throat contributes to the directional Landau-like emission threshold; a slow quadratic branch may drive the critical velocity to zero unless its coupling vanishes by symmetry. In a dissipative or mixed branch, the absence of an on-shell wave is not enough: low-frequency friction, order-relaxation lag, diffusion, and conversion drag must be calculated directly. The moving-throat stage succeeds only if one family provides positive ordinary response, acceptable low-energy universality, one electric-magnetic-radiative normalization, and bounded extra radiation and drag in the declared regime.

7.4 Two-throat interactions and finite-thickness crossover

The two-body program begins from two independently solved throats, not from two adjustable point sources. At large separation their measured source profiles must reproduce the realized far-field forces of Candidate V1. The solutions are then continued inward while tracking how each throat changes the other's core, support mode, transport, conversion, distributed porous-brane return halo, and reservoir data. This continuation tests whether far-field factorization and approximate additivity remain valid when the sources are no longer isolated.

Three separation regimes organize the calculation:

Regime	Dominant description	Required output
$R \gg H_{\text{br}}, R_{\text{throat}}$	Normalized far-field modes and solved one-throat residues	Recovery of gravity, odd electric, magnetic, and any allowed subleading forces with the frozen normalizations.
$R \sim H_{\text{br}}$	Full finite-thickness interface and parent-field response	Cross-interface form factors, even thickness forces, polarization, overlap of porous-brane return profiles, and departure from point-source factorization.
R comparable to the nonlinear core scale	Coupled parent-field two-core solution	Core deformation, support-mode phase and stability, topology, contact, short-range force, radiation, and possible merger or exclusion.

The pair classes must include same-species same-orientation throats, same-species opposite-orientation partners, and oppositely charged throats of different or composite species. The odd Coulomb channel, reflection-even thickness and material channels, gravity and distributed porous-brane return response, magnetic terms, and nonlinear core interaction must remain separately visible. A correction to the odd monopole coefficient cannot be used to hide an even or core force, and distributed porous-brane return-halo overlap must be allowed to change the signed effective gravity source rather than being absorbed into an absolute mass parameter.

The force construction depends on the solved branch. A genuine equilibrium pair may admit an ensemble-appropriate static energy. A conservative periodic pair may require fixed support-mode action, frequency, and relative phase or a cycle-averaged stress force. A maintained dissipative pair requires the complete stress, momentum-flux, conversion, and reservoir response unless a conservative potential is independently established. This proposal does not reproduce the full ensemble and Green-matrix program; those calculations remain in the canonical ledger and the focused opposite-orientation companion. The proposal-level requirement is that the physical force and its sign be derived from one well-defined pair problem rather than selected by changing the ensemble.

A successful crossover calculation connects the far-field observables to a regular finite-thickness interaction and identifies where the two-interface graph description ceases to be adequate. It must also reveal environmental and many-body corrections to charge, gravity, mass response, and source additivity. An unphysical singularity, arbitrary branch jump, loss of conservation, or need for a new short-range coefficient would reject the selected pair realization.

7.5 Projected coincidence, core contact, and annihilation

A brane observer records a projected position $\mathbf{x}_a$, whereas the nonlinear core occupies a finite region of the full space $\mathbf{X} = (\mathbf{x}, w)$. Oppositely oriented throats may therefore have the same projected position while remaining separated across the slab. This is not a small correction to point-particle language; it changes what must be calculated before annihilation can even be defined.

The central logical separation is

odd electric cancellation $\nRightarrow$ even material cancellation $\nRightarrow$ core contact $\nRightarrow$ annihilation.

For a complete same-species orientation pair, the leading odd source may cancel at projected coincidence while the even thickness disturbance, support fields, flow, conversion zones, and other orientation-independent structure add. If the throats are periodic, exact instantaneous cancellation may additionally depend on the relative support-mode phase supplied by the complete antiparticle map; cycle-averaged electric neutrality does not remove possible phase-dependent stress or sideband radiation.

While both slab interfaces remain graphs, a useful diagnostic separation is

$$d_w^{\text{core}} = H_{\text{br}} + 2h_t^{\text{tot}} - \ell_+^{\text{in}} - \ell_-^{\text{in}},$$

where the even thickness disturbance changes the local interface spacing and the inward reaches $\ell_{\pm}^{\text{in}}$ must come from the solved cores. A positive value permits projected overlap with separated cores; zero marks first contact within that description; a negative value signals that the separate-core graph ansatz has failed. The formula is a diagnostic, not a substitute for the nonlinear solution.

Once overhangs, necks, tubes, reconnection, pinch-off, or topology change occur, the interface variables must give way to the complete parent fields. Contact then requires a physical localized core diagnostic and a continuation rule that keeps the two cores identifiable from large separation until their closures first touch or their component topology changes. The result must be robust under reasonable variation of the diagnostic threshold. After merger, separate “plus” and “minus” cores may not be reconstructed by an analyst-chosen split of one total field. The detailed set definitions and numerical contact protocol remain in the opposite-orientation companion.

Contact still does not select the outcome. A same-species throat–antithroat pair may scatter, form a metastable neutral state, reconnect, annihilate into allowed propagating and reservoir modes, or leave a neutral remnant. Because its net oriented count can vanish, annihilation may be topologically permitted, but the dynamics and conservation laws must show an actual channel. By contrast, two oppositely charged objects of different internal or composite species may retain unmatched odd multipoles, even material structure, support modes, or topology. Electric neutrality must not force them into the same annihilation channel.

The decisive calculation is a full time-dependent collision evolution using the parent fields. It must close constituent number, momentum, and energy and, for a relaxational or mixed branch, entropy and internal reservoir accounting. It must measure outgoing transverse light, odd and even disturbances, longitudinal and bulk waves, conversion and reservoir excitation, and any surviving core. Failure of immediate annihilation is not automatically a model failure; it may establish a bound state, exclusion mechanism, or metastable channel. The particle–antiparticle sector fails if no conservation-respecting annihilation or neutralization channel exists for a complete same-species pair, or if the theory forces materially different opposite charges to annihilate solely because their leading electric monopoles cancel.

7.6 Localized throat drainage, distributed porous-brane return, and the reservoir fixed point

Each throat supplies a localized, one-way drain. Material crossing its mouth de-structures into bulk and cannot return through that throat. Return occurs separately when bulk material reorganizes into brane state across the porous brane, represented by the positive bulk-to-brane contribution to the signed projected exchange term S_{leak} . The local drain, gross order conversion, distributed return, and reservoir response therefore belong to one conservation problem but are not one mechanism or one rate.

Every local throat calculation depends on asymptotic data supplied by the wider medium. Denote the provisional environmental state by

$$\mathcal{B}_\infty = \{n_\infty, P_\infty, \mu_\infty, \chi_\infty, T_\infty, \text{distributed } S_{\text{leak}} \text{ return profile, reservoir state}\},$$

with only the variables appropriate to the selected dynamical branch retained. A local solution obtained at fixed $\mathcal{B}_\infty$ exports throughput, conversion, stress, momentum, energy, radiation, and possible entropy to the global bulk. Those outputs alter the reservoir and distributed return profile; the changed global state then alters the backpressure and other local boundary data without sending material back through the throat. The final closure is therefore a fixed-point problem rather than a last one-way correction.

The iteration proceeds conceptually as follows:

1. solve the supported local throat under provisional $\mathcal{B}_\infty$;
2. extract its constituent flux, order conversion, momentum and energy transfer, emitted disturbances, and branch-specific entropy production;
3. solve the global bulk transport and reservoir evolution sourced by the population of throats, together with the spatially distributed bulk-to-brane return contribution to S_{leak} ;
4. update the backpressure, return scale, environmental asymmetry, gravity source profile, and local boundary data;
5. repeat until the local and global solutions reproduce the same reservoir state, distributed S_{leak} return profile, throat geometry, throughput, conversion, and emitted fluxes.

The global problem must preserve the one-medium ontology. Drainage is not loss into an external void. Material that crosses a throat remains in the common bulk medium but does not return through that throat. Return is neither reverse throat flow nor material supplied by an inexhaustible battery; it is the separate distributed reorganization of bulk material into brane state across the porous brane. A conservative branch must close total constituent number, momentum, and energy. A relaxational or mixed branch must additionally track the internal heat, disorder, entropy, and reservoir variables into which resolved order and mechanical energy are transferred. Any maintained stationary throat or long-lived particle must identify where the required energy resides and how the reservoir changes over time.

The fixed point also completes several observables that cannot be closed locally. The distributed porous-brane S_{leak} return determines whether the gravity source has a usable local matching region, becomes screened at a return scale, overshoots, or changes sign. The reservoir state influences throat radius, support frequency, lifetime, conversion, drag, and the comparison of the two orientations. A nominally reflection-symmetric particle branch may acquire an environmental splitting if the two bulk-facing reservoirs are not symmetric; that effect must be calculated rather than folded into the intrinsic antiparticle map.

A viable fixed point is stable or sufficiently long lived under perturbations of both the local throat and the global reservoir. It must not require secular reservoir depletion, unlimited entropy export to an unmodeled bath, or continual retuning of boundary data. Several outcomes remain scientifically useful. A stable fixed point would close the particle and reservoir program for the tested regime. A slowly evolving solution could define a metastable particle sector with a calculated lifetime and environmental dependence. A failure to converge may reject the selected order-dynamics or distributed-return branch even if its isolated throat looks acceptable. If every allowed branch of the frozen candidate lacks a conservation-respecting local-global fixed point, Candidate V1 fails at the final closure stage.

The result of this chapter is therefore not a promise that the far-field analog automatically contains particles. It is a sequence of nonlinear existence, response, interaction, topology, and reservoir tests that replaces each provisional source with a solved object. Only a candidate that survives this sequence can claim one connected classical analog extending from the common brane and far field to stable particle-like throats and their global material cycle.

Verification, Milestones, and Possible Outcomes

The preceding chapters define the proposed ontology, the sector calculations, the common-medium test, and the later throat and reservoir program. This final chapter states what evidence permits any of those claims to advance and what conclusion is authorized at each stage. It does not reproduce the detailed artifact schemas, discrepancy taxonomy, numerical procedures, or revision records maintained in the derivation ledger. Its purpose is narrower: to keep the top-down proposal, the bottom-up calculations, and the scientific decision sequence attached to the same controlled candidate.

A result belongs to the current program only when the question, assumptions, branch, frozen inputs, calculation, evidence, and downstream use all refer to the same version. A recent file, a successful script run, or a plausible plot is not enough. Conversely, a negative or inconclusive result is not discarded merely because it interrupts the intended sequence. The verification process is part of the falsification test because it determines whether several apparent sector successes truly arise from one medium.

8.1 Proposal-to-ledger verification

The proposal and the derivation ledger are independent views of the same research program. The proposal works from observable behavior downward: it states what must be reproduced, which unwanted channels must be included, what the result would require from the common medium, and what would reject the branch. The derivation ledger works from the candidate equations upward: it records the selected branch and boundary data, performs the symbolic and numerical calculations, preserves the actual signs and extra modes, and tracks the inputs and outputs inherited by later steps.

These views should be compared through a concise proposal-to-ledger crosswalk. For every mandatory requirement, the crosswalk should identify the exact claim, the Candidate Parent System version and dynamical branch, the background and environment, the boundary or interaction ensemble, all frozen inputs, the valid scale and approximation regime, the ledger steps that test it, the human and executable evidence, the expected and observed result, the current decision, and the downstream dependencies. One requirement may need several calculations, and one calculation may support

several requirements; the mapping therefore must work in both directions.

The minimum completeness rule is straightforward. No mandatory proposal claim should lack either a current evidence path or an explicit open-status record. No promoted ledger result should lack a stated proposal purpose and claim boundary. No derived output should be used downstream without a dependency record, and no current pass should rely on a superseded candidate, an unstated ensemble, or a reopened upstream result.

This comparison is also where discrepancies become scientifically useful. The proposal is not rewritten automatically to match an unexpected output, and a complicated expression is not declared successful merely because the ledger produced it. A mismatch may reveal that the proposal asked the wrong physical question, that the derivation used different assumptions, that a convention or implementation is inconsistent, or that the candidate fails. Until the earliest cause is identified and the affected evidence is rerun, the claim remains open or reopened.

Current calculation status is therefore read from the derivation ledger, executable artifacts, and the independent Codex repository audit rather than from a dated status chapter in this proposal. The proposal records the enduring research contract; the audit records which parts of that contract presently have current evidence.

8.2 Required symbolic and numerical evidence

Every load-bearing claim begins with a human-readable derivation. It should state the governing equations and whether each is postulated or inherited, the field variables and units, the conservative, relaxational, or mixed branch, all constraints and symmetries, the boundary or reservoir data, the approximation order, and the conclusion the calculation is authorized to support. Intermediate steps must remain visible where signs, cancellations, singular limits, source normalizations, or branch choices affect the result. A necessary-condition calculation obtained by solving for a desired coefficient may be useful, but it must not be reported as a derivation of that coefficient from the candidate.

Where symbolic verification is feasible, SymPy and Mathematica should construct the central mathematics independently from the same controlled specification. They may share the symbol dictionary, assumptions, and expected invariants, but one implementation should not import the other’s final symbolic expression. Comparison should use canonical objects rather than visual similarity: equation residuals, characteristic polynomials, root multiplicities, invariant subspaces, projectors, ranks, principal minors, series coefficients, symmetry transforms, current divergences, and high-precision samples at regular and near-threshold points. Exact arithmetic and explicit assumptions should be retained as long as practical.

Agreement between the two systems supports the algebraic claim under the stated assumptions. It does not establish that the parent postulates are true, that the required nonlinear branch exists, or that the result is observationally acceptable. If one system cannot complete a calculation, the missing check may be replaced by an independent hand reduction, another mathematical method, a controlled high-precision residual, or an alternative numerical implementation, but the asymmetry must remain visible.

Numerical evidence is required for most existence, localization, stability, and closure claims. A stable

finite slab, guided or resonant modes, a localized photon candidate, a supported throat, moving and accelerating families, two-core contact, and the local-global fixed point cannot be established by a formal symbolic expression alone. The numerical record must retain the exact equations and baseline, domain and boundary conditions, discretization and solver, initial data or continuation seed, raw solution fields, residuals and constraint violations, and the observables used by the gate.

A gate-level numerical result also requires the refinements relevant to its physical classification. These may include grid or basis refinement, timestep and domain variation, outgoing-boundary or regulator variation, precision increase, continuation in both directions, multiple seeds, and comparison of independent solvers or discretizations at selected points. Conservation, supplied and dissipated power, radiation into every coupled branch, entropy and reservoir balance, and linear, Floquet, or retarded-response stability must be checked when the chosen branch requires them. Open, dissipative, or non-normal problems may additionally require complex poles, resonance widths, residues, passivity, and sensitivity to transient growth.

The evidence standard must match the mathematical problem. A static energy minimum cannot validate a maintained dissipative state. A conservative Floquet spectrum cannot close a mixed branch whose reservoir was omitted. A reduced envelope or imposed throat geometry is a precursor until it is continued to the parent fields. Failure of one solver or one seed is not a proof of nonexistence; one converged solution is not proof of uniqueness, robustness, or an acceptable family.

8.3 Milestones and decision gates

The gates follow the mathematical closure order, even when conditional work is scheduled in parallel. Each exit condition authorizes the next level of claim; it does not confer success on later stages. The lean program uses the sequence in table 8.1.

Table 8.1: Major verification and decision gates.

Gate	Milestone	Minimum exit condition
G0	Controlled research contract	Canonical definitions, proposal requirements, failure conditions, shared quantity identities, and the proposal-ledger crosswalk are current and mutually consistent.
G1	Light and far-field capability contracts	The light, gravity, electric, magnetic, radiation, mass-response, extra-mode, leakage, and drag programs have produced explicit common-medium requirements and named failure channels rather than separate tuned solutions.
G2	Common region and Candidate V1	The Medium Capability Matrix identifies a nonempty contract-feasible region, and one Candidate Parent System version freezes its fields, branch, equations, operator basis, coefficients, symmetries, source conventions, and boundary and reservoir data.
G3	Stable background and common far field	Candidate V1 produces a selected stable or sufficiently metastable finite brane, a complete acceptable constrained spectrum, and all mandatory declared light and far-field capabilities at the same current point or region.

Gate	Milestone	Minimum exit condition
G4	Supported throat and species basis	At least one self-consistent stable or sufficiently long-lived supported oriented throat exists; its source profiles replace the provisional sources; and the required orientation-partner, species, charge, and conservation tests are defined and passed to the declared scope.
G5	Moving and accelerating throat closure	The solved throat family yields positive ordinary passive and inertial response, acceptable low-energy universality, one electric–magnetic–radiative normalization, and bounded extra radiation, leakage, wake, and drag.
G6	Two-throat and contact closure	The same solved objects recover the far field, survive the finite-thickness crossover, and produce conservation-respecting contact, annihilation, exclusion, or neutral-binding outcomes with same-species partners distinguished from different species.
G7	Localized throat drainage, distributed return, and reservoir closure	The local one-way throat problem and the global bulk-reservoir problem, including distributed porous-brane return through the positive bulk-to-brane contribution to S_{leak} , converge to a stable or sufficiently long-lived common fixed point with material, momentum, energy, and, where applicable, entropy and internal-reservoir accounting closed.

The localized photon-packet track is a stronger light milestone that branches from the stable-background and complete-spectrum result. It may be pursued in parallel with later realization work, but it receives its own controlled existence, stability, leakage, and collision decision. Failure of that stronger track narrows the outcome to guided classical light unless the frozen candidate explicitly made a self-bound packet mandatory; it does not silently convert an unsolved packet into a passed light result.

A gate may be recorded as *pass*, *conditional pass*, *reopen*, *fail*, or *open/inconclusive*. A conditional pass must name the unresolved condition and the limited reason downstream exploratory work remains useful; it cannot conceal a known violation of a mandatory requirement. A fail means the current branch or candidate misses a frozen condition. An open result means that evidence, convergence, domain coverage, or the mathematical object itself is still insufficient to decide.

8.4 Revision and downstream reopening

A substantive upstream change is a scientific change, not a prose correction. If a field, constitutive term, coefficient, dynamical branch, source definition, calibration, boundary ensemble, or reservoir law changes, the candidate receives a new version and every actual dependent result is marked reopened until it is rederived. Derived outputs are frozen downstream inputs: a guided-light profile, electric eigenvector, throat residue, support frequency, or reservoir state cannot become adjustable again merely because a later sector prefers another value.

Reopening should follow real dependencies rather than trigger ceremonial reruns of unrelated work.

A revised shear law or slab-thickness mechanism reopens the background spectrum, guided light, electric response, characteristic speeds, and throat support. A revised oriented-current normalization reopens static charge, magnetism, transverse radiation, and charge universality. A revised order or reservoir branch reopens gravity, throughput, throat lifetime, inertia, drag, and global closure. A revised antiparticle map or physical core diagnostic reopens species, pair, contact, and annihilation claims without necessarily invalidating the homogeneous background spectrum.

Historical calculations remain valuable as failed branches, special limits, regression tests, or sources of new requirements, but they no longer count as current evidence for the revised candidate. A new mechanism introduced after a failure is permitted only as an explicit revision of the common parent system, with its consequences propagated through every affected sector. It may not be added privately to the calculation that exposed the problem.

8.5 Interpreting success, partial survival, and failure

The outcome of the program is layered. Every report should state the strongest coherent submodel that survives, the exact gate reached, the gate that failed or remains open, and the assumptions under which the conclusion holds.

Outcome	Authorized conclusion
Open or inconclusive	A required equation, branch, source, convergence study, acceptance bound, or domain test is missing or conflicting. Neither pass nor failure is justified.
Sector result	One named behavior, compatibility condition, or no-go statement is established within its assumptions. It need not establish any larger part of the ontology.
Partial analog	A coherent subset survives after a stronger claim fails or remains outside scope, such as guided classical light without a self-bound photon packet or a local gravity analog with a limited return regime.
Common far-field medium	One frozen candidate has a stable background and rederives all mandatory declared light and far-field sectors at one shared point or region. Particle existence and global closure remain unproved.
Supported particle sector	One or more stable or sufficiently long-lived throat branches exist with solved source profiles and the declared orientation, species, and charge structure. Moving, pair, or reservoir gates may still fail.
Moving and interacting particle sector	The candidate also closes ordinary response, universality, magnetism, radiation, drag, two-body crossover, contact, and the declared short-range outcomes. The global reservoir fixed point may remain open.
Full declared classical survival	One current parent system passes every mandatory gate through the local-global fixed point in the stated regimes, with all shared inputs frozen and all unwanted channels retained in the evidence.
Candidate failure	One named Candidate Parent System violates a mandatory gate or cannot close its own background, throat, interaction, or reservoir problem. Other admitted candidates are not thereby rejected.
Architecture failure	The mandatory intersection is shown to be empty throughout the admitted architecture, an essential mechanism is impossible across its permitted realizations, or success requires prohibited sector-specific retuning or an unsupported new mechanism.

Meaningful negative results include several important intermediate outcomes. Classical guided light may survive while every permitted self-binding mechanism for a free photon packet fails. Gravity and electricity may work in separate effective limits but have no common coefficient region. A stable common brane and far field may exist while no self-consistent supported throat does. Stable throats may exist while unavoidable drag, excess radiation, wrong-sign response, or nonuniversal charge or mass ratios reject the particle interpretation. Local particle physics may close while the global distributed-return and reservoir iteration has no conservation-respecting fixed point. Each result narrows the viable model and identifies where the one-medium program stops.

Full declared survival would be the strongest result available to this proposal, but its scope remains classical and conditional on the stated regimes. It would not establish that the real universe is literally this medium, reproduce exact general relativity or exact Maxwell theory, or supply quantum statistics, a universal quantum of action, or a complete particle spectrum. A later cosmological extension would remain a separate result unless deliberately made mandatory. The value of the program is therefore not limited to a final positive answer: a controlled partial analog, a precise

candidate failure, an empty common region, or a well-supported no-go result is also a successful scientific outcome because it is obtained without hiding dependencies or repairing the model after the fact.

Companion Documents

The proposal should end with a short controlled list of companion documents and repository locations, not a collection of new book-length appendices. The final assembly should include only the current canonical filenames and versions.

- Canonical ontology and closure ledger: **[TBD: final filename and version]**
- Guided-light and photon-soliton research plan: **[TBD: final filename and version]**
- Opposite-orientation throat and contact plan: **[TBD: final filename and version]**
- Cross-sector research burdens and compatibility gates: **[TBD: final filename and version]**
- Derivation-ledger root and repository revision: **[TBD: final path and commit]**

Scope control

Detailed glossary tables, epistemic rubrics, acceptance-budget ledgers, proposal-to-ledger cross-walks, parameter-dependency tables, failure registers, provenance tables, review checklists, and revision histories remain separate project-control artifacts unless a later reader-facing need justifies a small, specific appendix.